

Structure and a Little Training Beat the Gate: Adaptive-Compute Cascades and Trained Verifiers for Efficient, Accurate Medical VLMs

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Abstract

Deploying medical vision-language models (VLMs) is expensive: a 32B reasoning model spends ≈ 11 s and ≈ 6 kJ per question (batch-1), a 7B model ≈ 0.2 s. A **cascade** — answer cheaply, escalate only the hard cases — is the standard route to efficiency, but it requires *deciding* which cases to escalate and *what to do* once escalated. We report one hard negative, two positive levers, and a unifying explanation.

A hard negative — the luck floor. Over a *frozen* model, no *training-free* signal beats trivial baselines. A dozen escalation signals (confidence margin, MSP/Chow, entropy, Gini/DOCTOR, self-verification $P(\text{True})$, conformal prediction, learned routers, cross-model agreement, multi-resolution stability, semantic self-consistency) all hit the same **recoverability ceiling** ($\text{AUROC} \approx 0.5\text{--}0.69$): whether a cheap model's error is *fixable* is near-unpredictable from cheap features, because the two models fail together ($\phi = 0.372$; on the competent benchmarks $P(32\text{B wrong} \mid 7\text{B wrong}) = 0.584$). The same wall appears for *selection* — given N sampled answers, no training-free rule beats picking one at random (the luck floor) — and even for verifiable bounding boxes.

Two levers move the needle, neither of which is "a better gate." (1) **Structure (efficiency).** The *Adaptive-Compute Cascade* (ACC) routes across *compute configurations* of the same models — 7B-no-think $\rightarrow$ the 32B's **fast no-think mode** $\rightarrow$ the 32B's slow think mode — because reasoning *over-thinks* perception VQA (32B-no-think $\geq$ 32B-think on competent sets). At parity accuracy this cuts **latency 11.34 s $\rightarrow$ 2.27 s (-80%), FLOPs 100 % $\rightarrow$ 52 %, and energy $\approx 5\times$ (6.3 kJ $\rightarrow$ 1.2 kJ)** on ALL-6, never worse than the 7B on any benchmark; and a **faithful cross-family MCQ margin cascade** matches the 32B at **$-17 \dots -69$ % FLOPs**, with a reasoning **"think tier"** adding accuracy on the reasoning benchmark MMMU (**$+0.03 \dots +0.12$**) across three families. (2) **A little training (accuracy).** A small **trained outcome verifier** that scores $P(\text{correct} \mid \text{image, question, candidate})$ and selects best-of- N **breaks the luck floor and beats the strong 32B/38B on accuracy across three model families** — Lingshu, MedVLThinker, InternVL3 — on every dataset and a held-out out-of-distribution set (Lingshu pooled **0.421 vs 0.331**, MedVLThinker **0.344 vs 0.277**, InternVL3 **0.255 vs 0.218**), for free-text answers (recovering **35–49 %** of the oracle gap over two seeds) *and* structured bounding boxes (SLAKE organs **40–53 %**; the real **MS-CXR** chest-X-ray pathology benchmark **77–78 %**, a $5.6\times$ lift, bootstrap-significant). The verifier discriminates correct from incorrect candidates at **AUROC 0.924**, beats a zero-shot 32B verifier despite being $5\times$ smaller, and transfers zero-shot **across architectures** (a Lingshu-trained verifier lifts InternVL3).

The connective insight: training-free routing/selection over a frozen model is luck-floored; the two things that move the needle are *structure* (ACC) and *a little training* (the verifier). We additionally show the routing ceiling is partly a **benchmark artifact** — the same confidence signal jumps from $\text{AUROC} \approx 0.6$ on multiple-choice to ≈ 0.87 on open-ended free-text (a *discreteness* effect) — so medical-VLM cascades should be evaluated open-ended. We release ACC, the trained verifier, and the full negative-result characterization.

1. Introduction

A medical vision-language model (VLM) takes an image — a radiograph, a pathology slide, an endoscopy frame — together with a question, and returns an answer. The strongest medical VLMs today are large reasoning models that emit a long `<think>` trace before answering: they are accurate but slow and energy-hungry (for a 32B at batch 1, one answer costs ≈ 11 s and ≈ 6 kJ), whereas a 7B of the same family answers directly in ≈ 0.2 s. The standard route to efficiency is a **cascade**: run the cheap model first and escalate only the hard cases to the expensive one. A cascade's quality hinges on two decisions that are usually collapsed into one: the **gate** — *which* queries to escalate — and the **action** — *what* computation to run on escalation. This paper separates the two, characterizes the sharp limits of the first, and finds the real leverage in the second, plus a distinct training-based lever that improves accuracy rather than efficiency.

The gate is saturated — a "luck floor." A frozen confidence, agreement, or self-consistency signal can rank queries by how *likely the cheap model is wrong*. But what a cascade actually needs is different: whether *the expensive model will fix* the cheap model's error — its *recoverability*. We show recoverability is nearly unpredictable from any cheap signal. A dozen training-free signal families (margin, maximum softmax probability, entropy, Gini/DOCTOR, conformal scores, self-verification, a hidden-state probe) all cluster at the same ceiling, $\text{AUROC} \leq 0.69$, and a learned hidden-state probe (0.60) does not beat plain confidence (0.68). The mechanism is that the two models' errors are **correlated**: on the six-benchmark suite the 7B/32B recoverability rate is $\phi = 0.372$, and on the four benchmarks where the method is

competent, the strong model *also* misses **58.4%** of the cheap model's errors — $P(32B \text{ wrong} \mid 7B \text{ wrong}) = 0.584$. This is exactly the regime where post-hoc deferral theory (Jitkrittum et al., NeurIPS'23) predicts a wall, and no decision rule we tested beats a plain confidence threshold in a way that is simultaneously novel, real-efficiency-positive, and per-benchmark guardrail-safe. The same negative extends to *selection* (best-of-N over one model's own samples sits at a luck floor, below or barely above a random pick), to *actions* (cheap repairs recover a real but unharvestable 14% of errors the 32B also misses), to *cross-family peers*, and to the *language prior*. We call this unifying negative result the **luck floor** (§5).

Yet two levers give large, real gains — and neither is "a better gate."

Lever 1 — structure (efficiency). Instead of a better gate we change *what escalation runs*. The 32B has a *fast no-think mode* (≈ 0.34 s at batch 1) that, on perception VQA, is **as accurate as or better than** its slow think mode — reasoning *over-thinks* perception (SLAKE: 32B-no-think 0.849 vs 32B-think 0.764, +0.085; VQA-RAD: 0.853 vs 0.776, +0.077). Our **Adaptive-Compute Cascade (ACC)** inserts 32B-no-think as an intermediate tier between the cheap 7B and the slow 32B-think pass, gated by the agreement of the two no-think legs, so that the expensive think pass fires only on the $\sim 15\%$ reasoning residual. Because the cheap leg's latency is $\sim 80\times$ lower than a think pass, removing think calls dominates the savings: at parity accuracy on the six-benchmark suite (ALL-6), batch-1 latency drops **11.34 s $\rightarrow$ 2.27 s (-80%)**, FLOPs **100 $\rightarrow$ 52%**, energy **6318.8 $\rightarrow$ 1181.9 J ($\sim 5\times$)**, and the cascade is **never worse than always-7B on any benchmark** (guardrail = 0). Excluding the near-chance MedXpert set (ALL-5), latency drops **8.88 s $\rightarrow$ 0.44 s** and FLOPs to **24.9%**. Crucially, every training-free gate we swap in lands in the same 49–62% FLOPs band — confirming it is the *structure*, not the *gate*, that pays (§7–§8).

Lever 1, generalized — the faithful MCQ cascade and a reasoning "think tier." Reproduced under the faithful MedEvalKit protocol (our Lingshu-32B MMMU = 0.633 exactly matches the paper's 62.3), a plain 2-tier margin cascade matches the 32B at **$-17\ldots-69\%$ FLOPs** across three families and multiple benchmarks; the win magnitude tracks the 7B/32B accuracy gap, so we honestly report the keep-cheap and no-win cells too. Where a benchmark rewards reasoning, a strong-leg *think tier* adds accuracy for free: MMMU-Medical improves **+0.034** (Lingshu, 0.633 $\rightarrow$ 0.667), **+0.107** (MedVLThinker, 0.613 $\rightarrow$ 0.720), and **+0.120** (InternVL3, 0.633 $\rightarrow$ 0.753) (§11).

Lever 2 — a little training (accuracy). Routing *between* frozen models is capacity-bound, but a different question — *given N samples from one model, which is correct?* — admits a **trained** answer. A small LoRA outcome verifier that scores $P(\text{correct} \mid \text{image, question, candidate})$, used for best-of-N selection, **breaks the selection floor** where every training-free selector is stuck. It **beats the strong model on accuracy across three families and held-out OOD**: pooled, Lingshu **0.421 vs 0.331**, MedVLThinker **0.344 vs 0.277**, and InternVL3 **0.255 vs 0.218** — the last obtained by transferring the *Lingshu*-trained verifier onto InternVL3's answers, i.e. across model architectures. The signal is learnable but not zero-shot-surfaceable: a trained 7B verifier beats a *zero-shot* 32B verifier (0.403 vs 0.355), so task-training matters more than scale. The same principle recovers **40–78%** of the oracle gap on bounding boxes, including on the real MS-CXR chest-X-ray benchmark (bootstrap gain +0.191, 95% CI [+0.152, +0.232]). We keep the caveats visible: on the canonical same-split pooled comparison the verifier is **+0.039 [+0.010, +0.066]** over the 32B on one seed but a **-0.005 tie** on the other — so the honest claim against the 32B is *matches-to-modest-win* (mean +0.017), while the improvement *over training-free selection* is robust on both seeds (+0.066/+0.088), and the per-dataset spread is wide (gap-captured 15% on SLAKE up to 58% on hard sets) (§9–§10).

Open-ended evaluation matters. The pessimistic routing AUROC (~ 0.6) is partly an artifact of multiple-choice: a single A/B/C/D letter is maximally discrete, so a confidence margin carries little information. On open-ended free-text the *same* confidence signal reaches AUROC **~ 0.87** (pooled 0.846, 95% CI [0.830, 0.862]); this is a discreteness effect, not a length effect (token-F1 agrees). We therefore study selection and verification open-ended throughout, which is what makes Lever 2 possible (§6).

Honest scope. Several results are explicitly provisional and flagged as such: the 7th benchmark (OmniMedVQA, $\sim 89k$ questions) is still running; the InternVL3-38B MedXpert cells are not yet measured (a context-length cap gap, re-run queued); and some July latency measurements taken under GPU contention are uncitable pending a serial re-run (accuracy and FLOPs are unaffected).

Contributions.

ACC, a cascade over *compute configurations* (7B-no-think $\rightarrow$ 32B-no-think $\rightarrow$ 32B-think) that exploits reasoning's over-thinking of perception, with large *measured* batch-1 latency/energy/FLOPs savings at parity (ALL-6 -80% latency, FLOPs $\rightarrow 52\%$, energy $\sim 5\times$) and a per-benchmark safety guarantee, shown to be structure-driven rather than gate-driven, across five model families.

A **luck-floor characterization**: a dozen training-free escalation *and* selection signals, all capped at a recoverability ceiling (AUROC ≤ 0.69), explained by 7B/32B error correlation ($\phi = 0.372$; $P(32B \text{ wrong} \mid 7B \text{ wrong}) = 0.584$ on the competent-4), and extended to actions, cross-family peers, the language prior, and structured outputs.

The **open-ended ceiling-break**: the routing signal's weakness is largely an MCQ discreteness artifact (AUROC ~ 0.6 MCQ $\rightarrow \sim 0.87$ open-ended), motivating open-ended evaluation of medical-VLM cascades.

A **trained outcome verifier** for inference-time best-of-N that beats the strong 32B/38B on accuracy across **three families and held-out OOD**, transfers across architectures, and unifies **free-text answers and bounding-box grounding** (incl. real MS-CXR) — 2-seed and bootstrap-significant, a test-time-scaling method that lets a 7B match or exceed the 32B single pass, with the ties and per-dataset failures reported inline.

A **faithful MedEvalKit MCQ cascade** (−17...−69% FLOPs at reproduced-32B accuracy) plus a strong-leg **think tier** that adds reasoning accuracy where headroom exists (MMM $+0.03...+0.12$ across three families), generalizing the structural lever beyond the internal harness.

Honest novelty framing throughout: the agreement *gate* mechanism is shared with prior cascading work (ABC/CAR) and the verifier lineage descends from generative reward models (GenRM); our contribution is the *compute-configuration structure*, the trained-verifier *application and unification* over answers and boxes, and the negative-result *characterization* — not the underlying signals.

2. Related Work

Our two decisions — *which* queries to escalate (the gate) and *what* to run on escalation (the action) — sit at the intersection of several literatures: LLM/VLM cascades and query routing, selective prediction and conformal deferral, the theory of post-hoc deferral, and verifiers/reward models for best-of-N selection. We survey each below and state, inline and without hedging, exactly which pieces we adopt unchanged and which are new. The short version: our **gate** and our **verifier architecture** are each squarely inside an existing family — we do not claim either as novel — and the contribution is (i) the *compute-configuration* structure the gate controls, (ii) the *application and unification* of an outcome verifier across free-text answers and grounding boxes in medical VQA, and (iii) the empirical *characterization* (the luck floor, and the multiple-choice-vs-open-ended discreteness effect).

Efficient cascades and query routing. FrugalGPT popularized the cost-saving LLM cascade: run a cheap model first, and escalate to a larger one only when a learned post-generation confidence/benefit score is low. Agreement-based cascading (ABC) escalates on ensemble *disagreement* rather than a scalar confidence, and confidence-aware routing (CAR, the closest prior art to our structural method, arXiv 2505.15154) routes reasoning queries by a confidence signal. Our **agreement gate is, by construction, a member of the ABC family** — the tier-1→tier-2 "think" trigger fires when the large model's fast no-think answer disagrees with itself under an ϵ -margin tiebreak (§7), and we make no novelty claim for it. What is new is *what the gate switches between*: not two models, but three **compute configurations** of the same two models (7B-nothink → 32B-no-think → 32B-think), inserting the strong model's fast no-think mode as an intermediate tier because reasoning *over-thinks* perception VQA (§7–§8). A parallel line learns a *pre-generation* query router directly from image/text features (rather than from a generation's confidence); we benchmark this line and find it does not beat a post-generation confidence signal here (§5.2), and a SigLIP-feature router lands at chance for our complementarity-routing task.

Confidence, selective prediction, and conformal deferral. The training-free gate family we benchmark against the margin gate is the classical selective-prediction toolkit: maximum softmax probability (MSP / Chow's rule), predictive entropy, the Gini/DOCTOR score, energy-based scores, split-conformal thresholding and its cascade instantiations (CP-Router, LAC), and learning-to-defer (L2D) heads. In our ACC bake-off these all cluster together at the recoverability ceiling: on ALL-6 at 32B-think parity every training-free gate lands within a narrow band (FLOPs $\approx$ 49–62%, batch-1 latency $\approx$ 1.8–8.0 s), and no member of the family meaningfully separates from the deployed margin gate (§5.1, §8). In the open-ended verifier-gate bake-off the same holds against the verifier-confidence gate (§10).

Post-hoc deferral theory — the wall. Jitkrittum et al. (NeurIPS 2023) analyze *post-hoc* deferral: given a frozen small model, a rule that decides when to route to a frozen large model is bounded by how predictable the large model's *marginal correction* is, and they derive the Diff-Prob deferral score for that setting. This theory is a precise description of the ceiling we hit empirically: the question "will the strong model fix this error?" is only **AUROC ~ 0.6** from any cheap signal, and a faithful Diff-Prob gate reaches only **AUROC 0.708** — well below the verifier-confidence gate's **0.853** and worse on the deferral curve (ADC 0.3832 vs 0.3923) — because the strong model repairs only **6–10%** of the cheap model's errors (about **26%** on the subset where the strong model is genuinely competitive), and *which* ones is near-unlearnable (§5, §10). We therefore treat Jitkrittum's result as the *explanation* for our luck floor rather than a baseline to beat.

Verifiers and reward models for best-of-N. Generative verifiers (GenRM) recast reward modeling as next-token "is this correct?" prediction and use the resulting score for best-of-N selection in *text*; vision-language process-reward models rerank intermediate *reasoning steps*; and medical generator-verifier pipelines exist, but primarily for *data synthesis / filtering*, not inference-time answer selection. Our verifier is squarely in this lineage — it is a small LoRA outcome head $s_\phi(v, q, a) = P_\phi(\text{Yes} | v, q, a)$ trained with a BCE objective (§9), and we claim no architectural novelty for it. The contribution is the **application and unification**: a single *outcome* verifier used for inference-time best-of-N selection in **medical VQA**, trained and evaluated over **both free-text answers and bounding-box grounding** (with the box label $y=1[\text{IoU} \geq 0.3]$) — a combination we did not find in prior work (§9–§10). We position this explicitly as a constructive counter to **Verification Mirage [2605.10850]**, which concluded that self-verification *fails* in medical VQA and recommended retrieval rather than a trained verifier. That negative result is about *zero-shot self-verification*,

and our evidence is consistent with it: zero-shot self-verification is essentially uninformative (discrimination AUROC $\approx$ 0.5), and a zero-shot 32B verifier underperforms a *trained* 7B verifier (selection accuracy 0.355 vs 0.403). The difference is training: the same signal, once *learned*, reaches discrimination AUROC **0.924** and turns into an accuracy win over the strong 32B/38B across three model families and held-out OOD (§9–§10). So we read the Mirage result as scoped to the zero-shot regime, not as a bar against trained outcome verifiers.

Verifier-score-as-gate precedents. A recent thread uses a verifier/confidence score itself as the *deferral gate*: CCPS [2505.21772] scores answer reliability via input-perturbation (visual/consistency) stability; Self-REF [2410.13284] trains the model to emit calibrated self-reference confidence tokens for routing; and Kiyani et al. [2602.17633] give a conformal treatment of verifier scores for selective prediction. Our escalation gate — thresholding the trained verifier's confidence — is in this spirit, and we again claim no novelty for the *idea* of gating on a verifier score. **None of these is a trained outcome verifier for medical VQA selection**, and where they overlap our setting we benchmark them faithfully: a faithful CCPS input-perturbation visual-stability gate reaches only **AUROC $\sim$ 0.60** (0.604 pooled) and adds nothing when combined with verifier-confidence (**0.852** $\rightarrow$ **0.853**, i.e. within noise; §10). The verifier-confidence gate remains the best member of this family in our bake-off, and no *trained* gate beats it by more than +0.008 AUROC (§10).

Base families and the faithful evaluation protocol. Our three base families are Lingshu-7B/32B [2506.07044], MedVLThinker-7B/32B, and InternVL3-8B/38B (verifier base: Lingshu-7B; box-verifier base: Qwen2.5-VL-7B). We reproduce the Lingshu MedEvalKit evaluation protocol faithfully rather than through an internal harness; the protocol recipe and our reproduction-fidelity numbers are deferred to §4, and the faithful MCQ-cascade results that build on it are in §11.

Open-ended vs multiple-choice routing — the discreteness claim. Semantic-agreement cascades for text LLMs escalate on cross-model or cross-sample agreement, and are typically evaluated on open-ended generation. Our contribution to this line is the observation, which we did not find stated elsewhere, that the *routing* of a medical VLM is largely a **multiple-choice artifact**: the *same* confidence signal that is nearly useless on MCQ (routing AUROC $\sim$ 0.6) becomes strongly discriminative once the answer space is open-ended ($\sim$ 0.87), and we show this is driven by answer *discreteness*, not answer *length* (token-F1 agrees; §6). The practical corollary — that medical-VLM cascades and verifiers should be studied open-ended — motivates the selection and verification results in §6, §9, and §10.

3. Setup, Metrics, and Definitions

This section fixes the models, data pools, cost accounting, decision signals, and evaluation metrics used throughout. Everything downstream — the luck-floor characterization (§5), the open-ended ceiling-break (§6), and the three methods (§4, §9, §11) — is defined against the objects introduced here. Every number quoted below is a real measurement traced to a checkpoint; none is fabricated.

3.1 Models and families

The paper is organized around **three medical-VLM families**, each a small "cheap" leg and a large "strong" leg of the *same* architecture:

Family	Cheap leg	Strong leg
Lingshu	Lingshu-7B	Lingshu-32B
MedVLThinker (MVT)	MedVLThinker-7B	MedVLThinker-32B
InternVL3 (IV3)	InternVL3-8B	InternVL3-38B

Lingshu and MVT are the two families for which we have full efficiency instrumentation (batch-1 latency/energy, per-tier costs); InternVL3 is the third, cross-architecture family added to test whether the trained verifier and the faithful MCQ cascade *transfer* off the family they were tuned on. A single cross-cutting result depends on this: the outcome verifier is trained on Lingshu-7B answers only and then applied, unchanged, to InternVL3-38B candidate answers (cross-architecture verifier transfer; §10).

Two additional bases appear where a specific method needs them. The **outcome verifier** (§9–§10) is a LoRA adapter ($\sim$ 190 MB) on the *frozen* Lingshu-7B, trained on $\sim$ 6,000 judged (image, question, answer) triples at a positive rate of 0.194. The **box verifier** for grounding (§10.2) uses Qwen2.5-VL-7B as its base.

For the ACC generalization study (§8) we additionally run three more families that we do *not* otherwise instrument: **QoQ-Med** (7B/32B), **Chiron-o1** (2B/8B), and **MedGemma** (4B/27B). These are introduced only to show where the over-thinking premise behind ACC holds and where it fails (they are defined and used only in §8). Finally, the cross-family *complementarity* analysis in §5 uses two off-family peers (InternVL, Phi-3.5-V) as diversity sources; those are named where used.

A model quirk that matters for the cost accounting. The MVT and Lingshu strong legs can run in two modes: **no-think** (answer directly) or **think** (emit a `<think>...</think>` reasoning trace before the answer). The mode is not free-floating — these models only emit a `<think>` trace under one exact system prompt; without it they answer directly. The no-think vs think distinction is the axis ACC exploits (§4) and the think tier exploits (§11), so we treat "32B

no-think" and "32B think" as two distinct, separately-measured configurations of the same weights.

3.2 Benchmarks and evaluation pools

Multiple-choice benchmarks (7). PMC-VQA, SLAKE, VQA-RAD, PathVQA, MMMU-medical, MedXpertQA-MM, and OmniMedVQA. The core MCQ eval set is **8,220** samples across the first six; MedXpertQA-MM contributes two splits (Reasoning and Understanding), so per-split tables show seven columns. **OmniMedVQA is provisional** — the Open-access split is 88,996 QA pairs (of which ~57k, i.e. 64%, are RadImageNet: 56,697 items; 4-option MCQ, exact-match), and the Lingshu paper evaluates the full set, so faithfulness demands the full 89k rather than a sample. The full-set run was still executing at the time of writing (its parser bug — a `KeyError` over 42 sub-datasets — was fixed and the 89k run restarted), so no cascade result over OmniMedVQA is reported here; we carry it as in-progress.

We name three pools used repeatedly:

- **ALL-6** — all six MCQ benchmarks (n = 8,220; MedXpert split into R and U for per-split tables).
- **ALL-5** — ALL-6 minus MedXpertQA-MM. MedXpert is excluded from the ALL-5/headline claims because both legs sit near chance on it (e.g., MVT 7B/32B-think MedXpert-R 0.225 / 0.326, U 0.256 / 0.385).
- **COMPETENT-4** — SLAKE, VQA-RAD, PathVQA, PMC-VQA (n = 6,050): the sets on which both legs are competent and on which the ACC over-thinking premise (§4) is meant to hold.

Open-ended sets. For the open-ended studies (§6, §9-§10) we use free-text versions of SLAKE, VQA-RAD, PathVQA, and Kvasir-VQA-x1, plus **RadImageNet-VQA** as a held-out transfer/OOD set (a verifier trained on the first four is applied to RadImageNet zero-shot). Open-ended answers are graded by an LLM judge (§3.6), not by string match.

Grounding sets. For structured outputs (§10.2) we use SLAKE organ bounding boxes (n = 487) and **MS-CXR**, a real chest-X-ray phrase-grounding benchmark from PhysioNet (evaluation n = 435). A predicted box is scored correct iff its IoU with the reference exceeds $\theta = 0.3$ (§3.5).

3.3 The faithful MedEvalKit protocol (and why the internal harness is not it)

MCQ accuracies for the faithful cascade (§11) and the reproduction-fidelity check below are produced with **MedEvalKit**, run under a specific recipe so that our per-model numbers reproduce the published ones. The recipe: an isolated `medeval_venv`; the `Qwen2_5_VL` wrapper pointed at the family's weights; `datasets_path=hf`; `use_vllm` on; `TORCHDYNAMO_DISABLE` set (to avoid a compile-path crash); `use_llm_judge=False` for the MCQ (exact-match) halves; and `vLLM` pinned to **0.9.0.1**. We flag explicitly that our internal NGC labeling harness is **not** faithful for this purpose — it produces systematically different scores (e.g., a secondary NGC Lingshu-32B run gives SLAKE 89.4 / PMC 64.0 / MMMU 62.4, which do not track the paper the way the MedEvalKit run does) — so every "faithful" number in §11 comes from the MedEvalKit recipe above, and the NGC numbers are never used for a headline claim.

Reproduction fidelity. Under this protocol our MedEvalKit run reproduces the Lingshu paper on the anchor benchmarks (percentages; "ours" vs the paper's Table 6):

Benchmark	Lingshu-7B ours	Lingshu-7B paper	Lingshu-32B ours	Lingshu-32B paper
MMMU-Med	80.0	54.0	63.3	62.3
SLAKE-closed	82.5	83.1	85.9	89.2
PMC-VQA	54.3	56.3	55.2	57.9
MedXpertQA-MM	26.2	26.7	30.6	25.7
VQA-RAD-closed	78.1	67.9	85.3	76.5

The clean anchor is **MMMU-Med on the 32B: 63.3 vs paper 62.3 — an essentially exact match** on the same 150-question set. The 7B MMMU row is an **outlier by +26 points** (80.0 vs 54.0); this inflation is **Lingshu-7B-specific**, not a bug in our harness — MedVLThinker-7B on the same eval scores a normal **0.533** (below its own 32B), so we exclude the Lingshu-7B MMMU cell from all claims (this is why MMMU is "keep-cheap" for Lingshu in §11). Two honest caveats on the rest of the table: (i) SLAKE-closed and VQA-RAD-closed are the *closed* (MCQ) halves only, compared against the paper's *full-set* numbers, so they are not strictly like-for-like (closed halves are typically easier, which is why VQA-RAD-closed 85.3 sits above the full-set 76.5); (ii) the paper reference for MedXpertQA-MM-32B is reported as 25.7 in our reproduction log but as 30.9 in the paper's Table 6 — we record both and do not lean on that cell. The **open-ended halves** are reproduced separately via the Claude-Sonnet-5 judge (§3.6): VQA-RAD-32B = 74.1% (paper 76.5) and SLAKE-32B = 85.0% (paper 89.2), close but slightly under — consistent with a stricter judge on the open halves.

3.4 Cost model

One model forward pass over a query costs

F = 2 · N · (P + G) FLOPs,

(1a)

where N is the parameter count, P the prompt length in tokens (including vision tokens), and G the number of generated tokens; the factor 2 is the standard two-FLOP-per-parameter-per-token count and covers both prefill (P) and decode (G). We use $N_7 = 7.6\text{e}9$ for the 7B/8B cheap legs and $N_{32} = 33.0\text{e}9$ for the 32B/38B strong legs. For a cascade with per-tier cost c (in any of FLOPs, seconds, or joules) and escalation probabilities e_0 (fraction passing tier 0) and e_1 (fraction reaching the think tier), total cost is

$$C = c_{T0} + e_0 \cdot c_{T1} + e_1 \cdot c_{T2}. \quad (1)$$

We report **FLOPs%** (equivalently "backbone%") = $\Sigma(\text{cascade FLOPs}) / \Sigma(\text{always-32B-think FLOPs})$; 100% is the always-think strong model. **Latency** and **energy** use the same Eq. (1) with per-tier costs from real **batch-1** measurements; energy is NVML-integrated, $E = \Sigma (P_k + P_{k+1})/2 \cdot \Delta t$ over the power trace. The think-tier latency scales linearly with the number of generated tokens ($R^2 = 0.99$ for the fit), which is what makes the think tier so expensive: avoiding a think call avoids a variable, token-proportional cost, whereas the no-think tiers are near-constant (they decode ~ 2 tokens). *(All ACC efficiency numbers use the native batch-1 cost methodology of the canonical master_data.csv; an earlier co-resident rt_cascade estimate, which put the think leg at ~ 28 s rather than the batch-1 ~ 11.3 s, is superseded and not used for any headline.)*

Best-of-N cost. For the verifier method (§9–§10) each of the K sampled candidates costs one 7B generate, and selecting among them costs one 7B verify per candidate, so a best-of- K pass costs about $2K$ 7B-forwards for $K \geq 2$ ($K = 1$ needs no selection). In 7B-forward-equivalents:

Best-of-K	K = 1	K = 2	K = 4	K = 8
FLOP cost ($\times 7\text{B-fwd}$)	1 $\times$	4 $\times$	8 $\times$	16 $\times$

Against this, one always-strong (32B) forward costs **4.57** 7B-forward-equivalents, so the break-even is $2N < 4.57 \Rightarrow N \leq 2$: a best-of-2 verifier pass is cheaper than a single 32B forward, which is why the deployable open-ended recommendation is $N = 2$ (§10). Batch-1 latency does not follow the FLOP ratio — at batch-1 the 32B is only $\sim 1.9\times$ the 7B in latency and $\sim 2.8\times$ in energy (not $4.6\times$), because prefill and fixed overheads dominate.

A measurement caveat we carry honestly. Some July latency numbers — specifically the InternVL3 family and all PathVQA rows — were measured under GPU contention and are therefore **not citable** as batch-1 latencies; their accuracy and FLOPs are unaffected, and clean serial re-runs are queued. Where a latency is contended we say so rather than quoting it.

3.5 Confidence signals and the gate

Every gate in the paper thresholds a scalar computed from a greedy decode over the candidate answer tokens with logprobs $\pi_1 \geq \pi_2 \geq \dots$ and softmax probabilities $p_i = e^{\pi_i} / \sum_j e^{\pi_j}$:

- **margin** $m = \pi_1 - \pi_2$ (the deployed signal),
- **MSP** (maximum softmax probability) p_1 ,
- **entropy** $-\sum_i p_i \ln p_i$,
- **Gini** $1 - \sum_i p_i^2$.

A confidence gate escalates iff $m < \tau$ (low margin $\Rightarrow$ escalate). The threshold τ is fit on a held-out calibration split (PMC-VQA-train for MCQ; a cross-fit split for the open-ended verifier gate) and frozen; §5 and §11 report where a *trained* gate is compared against thresholding these raw signals.

3.6 The trained outcome verifier (definition)

The verifier is the one component of the paper that is *trained*. Given an image v , a question q , and a candidate answer a , it emits a scalar score

$$s_\phi(v, q, a) = P_\phi(\text{Yes} \mid v, q, a) = \text{softmax}(z)_{\text{Yes}}, \quad (4)$$

where z are the logits of a two-way (Yes/No) head on the frozen base (Lingshu-7B for free text, Qwen2.5-VL-7B for boxes) plus a LoRA adapter with parameters ϕ . It is trained with a binary cross-entropy loss against a correctness label y :

$$L(\phi) = -E_{\{(v,q,a,y)\}} [y \cdot \log s_\phi(v,q,a) + (1-y) \cdot \log(1 - s_\phi(v,q,a))], \quad (5)$$

where the label is $y = 1[a \text{ is correct}]$. For free-text answers correctness is the LLM-judge verdict (§3.6.1); for boxes it is $y = 1[\text{IoU}(a, \text{gold}) \geq \theta]$ with $\theta = 0.3$. At inference the verifier performs **best-of-N selection** over N sampled candidates,

$$\hat{a} = \text{argmax}_{\{i \leq N\}} s_\phi(v, q, a_i).$$

Note the verifier only has to *discriminate* good from bad candidates, not *generate* them — a strictly easier task than answering, which is the reason a small trained verifier can beat a much larger generator (\$9-\$10). The mechanism, non-circularity argument, and results are in §9-§10; here we fix only the definition.

3.6.1 Evaluation metrics

- **Accuracy** = exact-match (MCQ) or LLM-judge correctness (open-ended).
- **AUROC** of a score s for a binary correctness label y is the rank statistic $AUROC = P(s(\text{positive}) > s(\text{negative}))$ — the probability the score ranks a correct instance above an incorrect one.
- **Oracle@N** = $E_x[\max_{\{i \leq N\}} 1(a_i \text{ correct})]$: the accuracy of an oracle that picks a correct sample whenever one of the N candidates is correct. It upper-bounds any selector.
- **Gap captured** by a selector with accuracy $\text{acc}(\hat{a})$, relative to the greedy baseline and the oracle@N ceiling:

$$\text{frac} = (\text{acc}(\hat{a}) - \text{greedy}) / (\text{oracle@N} - \text{greedy}). \quad (2)$$

- **Guard** (safety) = the average number of benchmarks, per seed, on which a cascade falls *below* its own always-cheap (7B) baseline; 0 means the cascade is never worse than the cheap model.
- **IoU** is the standard box overlap; a grounding prediction is correct iff $\text{IoU} \geq \theta$, $\theta = 0.3$.

3.6.2 LLM-judge validation

Because open-ended accuracy depends on a judge, we validate the judge before trusting any open-ended number. The primary automated grader is **MedVLThinker-32B**; a **Claude-Sonnet-5** judge is used as an independent cross-check (and to reproduce the open-ended paper halves in §3.3). Three checks support the judge:

1. **Exact-match anchor.** On candidate pairs where the answer string equals the gold string ($n = 1,277$), the MVT-32B judge returns Yes **100.0%** of the time. Containment cases ($\text{gold} \subset \text{answer}$, $n = 2,423$) are judged Yes 82.5%; zero-word-overlap cases ($n = 14,320$) are judged Yes only 6.3%, and spot-checks confirm those are legitimate synonyms rather than judge errors.
2. **Inter-judge agreement.** A second 32B judge (Lingshu-32B vs MVT-32B) agrees at Cohen's κ in the **0.85-0.96** range:

Set	Agreement	Cohen's κ
VQA-RAD	0.984	0.962
Kvasir	0.958	0.849

3. **Independent frontier judge.** The Claude-Sonnet-5 judge, run over 1,257 SLAKE and 200 VQA-RAD open pairs, again shows a **100% exact-match anchor** ($n = 80/1005$ exact pairs, all judged Yes); its 21% zero-word-overlap judged-Yes rate was spot-checked to be entirely legitimate (synonyms and bilingual Chinese answers). Under this judge the open halves land at SLAKE-open 0.844 and VQA-RAD-open 0.600, reproducing the paper full-set numbers within a few points (§3.3).

We treat $\kappa \geq 0.85$ plus a 100% exact-match anchor as sufficient to use the judge as the open-ended accuracy oracle throughout §6, §9, and §10.

3.7 Peak VRAM

Peak GPU memory at batch-1 for each family and configuration (GB), used to size the two-GPU serving setup and to place each configuration on hardware:

Family (small / big)	Small leg	Big, no-think	Big, think
MedVLThinker, Lingshu, QoQ-Med (7B / 32B)	71.45	152.82	154.08
Chiron-o1 (2B / 8B)	53.46	74.16	74.17
MedGemma (4B / 27B)	52.35	155.52	155.52

For the 7B/32B families the strong leg exceeds a single 80 GB device, so the cascade co-resides the cheap leg and the strong leg on separate GPUs; think vs no-think adds essentially nothing to peak VRAM (154.08 vs 152.82 GB for the 32B) — the think-tier cost is entirely time and energy, not memory, which is exactly why ACC (\$4) trades *think calls* rather than *residency* for its savings.

4. The Luck Floor

This section states the paper's unifying negative and the reason the two positive levers (a structural cascade and a small trained verifier) are not "a better gate." The claim is sharp and holds across every axis we tested:

Over a frozen model, no training-free decision rule beats trivial baselines — for the gate (which query to escalate), the action (what to run on escalation), or selection (which of N sampled answers to keep).

We call this the **luck floor**. For the gate it is a *recoverability ceiling*: the best cheap signal predicts "will the strong model fix this error?" at only AUROC ~ 0.6 , because the two models fail together. For selection it is a *sampling floor*: a correct answer is often present among N samples, but the model cannot identify it, so every training-free selector ties or trails a random pick. The five sub-results below (a-e) are five faces of the same phenomenon. Everything positive in this paper is a way *around* the floor, not through it: the structural cascade (Method I) exploits that reasoning over-thinks perception rather than trying to route better, and the trained verifier (Method II) adds a *little supervision* precisely where training-free selection is floored. Section 5 (the open-ended ceiling-break) shows that the pessimistic ~ 0.6 gate AUROC is *partly* a multiple-choice artifact — but even in the favorable open-ended regime, training-free selection stays floored, which is what makes the trained verifier necessary rather than optional.

Unless noted, gate/action numbers are the MedVlThinker 7B/32B pair; selection numbers use Lingshu-7B generations judged by a fixed MedVlThinker-32B judge. ALL-6 = all six MCQ benchmarks; COMPETENT-4 = {PMC-VQA, SLAKE, VQA-RAD, PathVQA} (MMMU and near-chance MedXpert excluded).

4.1 The gate is saturated

A cascade gate does not need to know whether the *cheap* model is wrong; it needs **recoverability** — whether the *strong* model will fix that specific error. Cheap-model-wrongness and recoverability are different targets, and the second is far harder. Across **12 signal families** — margin, MSP/Chow, entropy, Gini/DOCTOR, energy, a hidden-state probe, self-verification P(True), conformal/CP-Router, a learned GBM router, cross-model agreement, multi-resolution stability, and semantic self-consistency — recoverability AUROC sits at **0.5-0.69**. The learned and hidden-state signals do not escape it: a hidden-state probe reaches only **0.60** versus plain confidence **0.68**, and a cross-validated "benefit predictor" trained to forecast the escalation gain tops out at **74.7%** on ALL-6 — *no gain* over plain confidence (73.7%). Even the *easier* target of predicting 7B-correctness is only mildly learnable (margin AUROC ~ 0.70 -0.75; self-verification P(True) 0.65-0.70, below margin), while the target a cascade actually needs — recoverability — is only ~ 0.6 from any cheap signal.

The cause is **error correlation: the two models fail together**. Decomposing every question by the joint (7B, 32B) right/wrong outcome:

outcome	ALL-6	COMPETENT-4
beneficial (7B wrong, 32B right)	17.6%	15.7%
futile (7B wrong, 32B wrong)	29.8%	22.0%
harmful (7B right, 32B wrong)	13.0%	13.4%
recoverability P(32B right 7B wrong)	37.2%	41.6%
$\Rightarrow$ share of the 7B's errors that are futile	62.8%	58.4%

So $P(32B \text{ wrong} \mid 7B \text{ wrong}) = 0.584$ on COMPETENT-4 ($\phi = 0.372$ on ALL-6, ϕ being the correlation of the two models' right/wrong outcomes): **most of what the small model gets wrong, the big model also gets wrong**. Recoverability rises only weakly with 7B uncertainty (43% $\rightarrow$ 28% across margin quintiles) and caps around 43-50% even for the most uncertain questions, so no confidence threshold can concentrate escalations onto recoverable errors. The consequence is visible in the deployed gate's own behaviour: of the margin gate's escalations, only **22.5%** are beneficial and **15.2%** are actively *harmful* (the 7B was right and the 32B is wrong). On the MCQ side the same wall appears directly — on COMPETENT-4 the 7B (cap320) scores 0.622 and the 32B-think 0.645 (gap +0.023); among the 2,286 questions the 7B gets wrong, the 32B is right only 41.6% of the time, and the margin ranks *which* of those are recoverable at AUROC **0.578**.

The headroom is real but unreachable. An **outcome-oracle** with perfect recoverability knowledge would reach 32B-parity using only 11.2% on the same backbone-compute metric at 4.6% escalation, versus the deployed gate's 73.6% at 63% escalation — a **62-point gap** that no training-free gate closes. This is not a failure of the particular signals we tried; it matches theory. Jitkrittum et al. (*When does confidence-based cascade deferral suffice?*, NeurIPS 2023) prove that the optimal deferral rule requires *both* models' confidence and that confidence-only deferral is fundamentally limited — exactly the wall we measure on medical VLMs. A learned "defer" meta-router that predicts escalation *gain* rather than cheap-model correctness (a VADR-style rule) is likewise not better and is not novel relative to prior confidence-deferral work; its one new claim — that the gain is learnable — fails.

4.2 Cross-family complementarity is real but unexploitable

Independently-trained VLMs make *different* mistakes, so their **oracle union** has large headroom: $\text{union}(7B \mid \text{InternVL}) = 0.753$ and $\text{union}(7B \mid \text{InternVL} \mid \text{Phi}) = 0.801$, versus always-32B 0.645. But that headroom is not addressable with cheap features. A learned router over frozen peer signals captures none of it — the best learned router reaches **0.621** $\approx$ **always-7B 0.622** — and a SigLIP image+text router that sees the raw question and image is at chance (**AUROC 0.50**). The complementarity exists; a training-free router simply cannot tell, per-question, *which* peer to trust.

4.3 The cheap model leans on a language prior

Part of what looks like "visual" difficulty is not visual at all. **56.9%** of the 7B's medical-VQA answers are *unchanged when the image is blanked* (blank-image probe), and questions the model is *image-sensitive* on versus *image-insensitive* on have nearly equal accuracy (**0.620 vs 0.625**). Much medical "VQA" is answerable from the text prior alone, which caps how much any better *visual* routing signal could buy — a large fraction of the questions carry no routable visual difficulty in the first place.

4.4 The action axis is capacity-bound, not action-bound

If the gate cannot pick recoverable errors, perhaps the *action* — what we do on escalation — can be made cheaper than "call the 32B." It cannot. Decomposing the 7B's errors on COMPETENT-4 (n = 6,050; error set = 2,286) by repair route:

repair route	share of the 7B's errors recovered
SCALE up (call the 32B)	41.6%
cheap 7B repair (look-closer OR think)	33.7%
multi-action oracle union (ceiling)	56.0%
cheap 7B catches what the 32B misses	14.3% (11-17% per benchmark)
unrecoverable by anything	44.0%

The genuinely novel observation is the fourth row: cheap same-model repairs recover **14.3%** of errors the 32B *also* misses — a real complementary signal, stable at 11-17% across benchmarks. But it is **unharvestable**: the repairs break as many answers as they fix (per-view accuracy is noise-limited — cap320 0.622, full-res 0.621, think 0.607 — versus 0.645 when the 32B is called), so a confidence-gated repair ladder *loses* at 32B-parity (**43% vs 39%** compute for the two-rung baseline). Max-confidence selection across five resolutions plus think lands at 0.608-0.622 and majority vote across views at 0.624-0.626 — i.e. no cheap combination of the small model's own views clears its single-view accuracy by a useful margin. The 32B's edge is *capacity*, not a cheap transform the 7B could apply to itself.

4.5 Selection is luck-floored (the headline negative)

The cleanest and most surprising face of the floor. Sample one open-ended model **N = 8** times: a correct answer is frequently *present* (large oracle gap — SLAKE-open greedy 0.730 → oracle@8 0.879, headroom +0.149; PathVQA-open greedy 0.343 → oracle@8 0.517, headroom +0.174), yet the model cannot say *which* of the eight it is. Every training-free selector ties or trails a **random pick** (0.720):

selector (SLAKE-open, Lingshu-7B, n = 645, MedVLThinker-32B judge)	acc	vs random	% of gap-above-random captured
random pick (mean sample accuracy)	0.720	—	—
self-verify P(Yes) argmax (7B)	0.715	−0.005	worse than random
self-consistency majority	0.736	+0.016	10%
32B pointwise verify, argmax	0.746	+0.026	16%
32B listwise select	0.758	+0.038	24% (best training-free)
learned fusion [self, 32B] (5-fold CV)	0.743	+0.023	14%
candidate-conditioned synthesis (32B primed with the 7B candidates)	0.774	—	(−0.045 vs free-gen)
32B free-gen single pass (SOTA bar)	0.819	—	—
oracle@8 (luck ceiling)	0.879	+0.159	100%

Even the best training-free selector (32B *listwise*) captures only **24%** of the gap above random, and *none* beats the 32B's own single free-generation pass (0.819); handing the 32B the 7B's candidates to *synthesize* an answer actually **backfires** (0.774 vs 0.819, −0.045). A **majority trap** explains the mechanism: the correct answer is a *minority* vote in **74-90%** of recoverable questions (mean ≈ 1.5 of 8 votes), so any consistency- or agreement-based selector is systematically pointed at the wrong answer. Two further training-free ideas fail in the same way. Cross-family agreement is correlated with correctness — P(correct | MedVLThinker-7B and Lingshu-7B agree) = 0.819 versus 0.649 on disagreement — but it buys nothing at the answer level: the agreement-consensus accuracy (0.730) merely equals Lingshu alone, and when the weaker model *dissents* it is right only 28.9% of the time. Few-shot in-context prompting *hurts* (PathVQA 0.343 → 0.203, −0.140; SLAKE 0.730 → 0.705, −0.025). The gap is **sampling luck, not latent knowledge the selector can surface for free**.

The same floor holds for **verifiable bounding boxes** with a competent grounder (Qwen2.5-VL-7B, IoU ≥ 0.3). Spatial self-consistency — choosing the medoid box across N samples — sits *at or below* the luck floor: on SLAKE organ boxes the medoid selector scores **0.164 vs greedy 0.197** (seed 0), i.e. *below* simply taking the first sample, and a zero-shot box verifier is likewise floored (0.177 vs greedy 0.199). Structured, geometrically-verifiable outputs do not escape training-free selection's limit any more than free text does.

Why this matters for the rest of the paper. The floor is not a counsel of despair; it is a map of where effort pays. It rules out a better *gate* over a frozen pair (§4.1–4.3), a cheaper *action* than capacity (§4.4), and any *training-free* selection rule (§4.5). What it leaves open is exactly the two levers this paper develops: a cascade that changes *what compute configuration* runs rather than routing better (Method I), and a small *trained* outcome verifier that supplies the one thing training-free selection lacks — a learned, per-answer correctness signal — which is the only thing shown to lift selection above the luck floor for both free-text answers and boxes (Method II).

5. The Open-Ended Ceiling-Break

The preceding luck-floor section delivered a deliberately pessimistic verdict: over a *frozen* model, no training-free gate reliably beats trivial baselines, with routing and recoverability AUROCs clustered at roughly 0.5–0.69. Taken at face value, that result reads as "medical-VQA cascades cannot be routed." This section shows that a large part of that pessimism is an **artifact of the multiple-choice answer format**, not an intrinsic property of medical VQA. The *same* confidence signal that looks nearly useless on multiple choice becomes strongly discriminative the moment the answer is free text: the AUROC for detecting a cheap-model error jumps from ≈ 0.6 on MCQ to ≈ 0.87 open-ended. Two things follow, one methodological and one that sets up the rest of the paper: (i) confidence-gated cascades *do* work open-ended, and (ii) the right regime in which to study selection and verification is open-ended — which is exactly where the trained verifier of Method II operates.

5.1 Why multiple choice is pessimistic: a discreteness argument

On a four-way MCQ, the model's output is a single letter A/B/C/D — a *maximally discrete*, degenerate target. Two consequences make confidence a weak correctness signal there. First, the target is coarse enough that a model can be "right for the wrong reasons" (a lucky letter) or "confidently wrong" with no surface trace, so the calibration of a top-1/top-2 margin over four symbols carries little information about actual correctness. Second, the base rate of chance agreement is high (a coin can land on the gold letter one time in four), which flattens any confidence-vs-correctness relationship. The net effect is the MCQ ceiling documented in the luck floor: whether the strong model will fix a cheap error is only ≈ 0.6 AUROC from any cheap signal, and even predicting cheap-model *correctness* tops out in the same neighborhood. **The claim of this section is that this ceiling is partly a measurement artifact of target discreteness, and that it lifts under free-text answers.** (The discreteness framing is our own; prior open-ended agreement work does not make the MCQ-vs-open-ended distinction.)

5.2 The ceiling-break: routing AUROC ≈ 0.6 (MCQ) $\rightarrow \approx 0.87$ (open-ended)

We re-run the same cheap-model confidence signal — the Lingshu-7B top-1/top-2 margin, escalating to Lingshu-32B — on the open-ended free-text sets (SLAKE, VQA-RAD, PathVQA), grading correctness with the validated LLM judge (§ Setup). The routing signal (predicting *cheap-model-wrong*, i.e. the event that should trigger escalation) rises to **AUROC ≈ 0.87** : pooled over the three sets, **0.846, 95% CI [0.830, 0.862]**, and **0.866** for the Lingshu-7B→32B cheap-model confidence on the $n = 845$ pooled gate-hunt slice. The bootstrap CI lies entirely above the MCQ ceiling — no resample falls at or below 0.6 — so the gap between the two regimes is not sampling noise. Per set, two prediction targets behave very differently:

Free-text set (Lingshu-7B $\rightarrow$ 32B)	Cheap-wrong AUROC	Recoverability AUROC
SLAKE	0.889	0.847
VQA-RAD	0.717	0.579
PathVQA	0.797	0.517
Pooled	0.846 [0.830, 0.862]	0.591

Cheap-wrong = predicting whether the cheap 7B answer is incorrect (the routing/escalation target). *Recoverability* = predicting whether the 32B will actually fix that error. The MCQ comparator for both is ≈ 0.6 (the routing/recoverability AUROCs of the luck-floor section).

Two honest readings of this table. **(a) The routing signal genuinely breaks the MCQ ceiling.** The *cheap-wrong* column — the quantity a gate needs — is the one that jumps (pooled 0.846; SLAKE 0.889, PathVQA 0.797, and even the hardest set, VQA-RAD, at 0.717 is above the MCQ ceiling). This is the headline: a plain confidence gate can tell when the 7B is wrong on open-ended answers far better than the MCQ numbers implied. **(b) Recoverability does not break.** The *recoverability* column — "will escalating actually help?" — stays near the luck floor once pooled (0.591), matching the recoverability wall of the previous section. In other words, the format artifact inflates the difficulty of *detecting cheap errors*, but it does not manufacture recoverable structure that was not there: whether the strong model repairs a given error remains largely unlearnable from cheap signals, consistent with § from the luck floor. Both

facts are kept intact here — the open-ended regime helps the gate, not the underlying complementarity.

5.3 It is discreteness, not answer length

A natural confound is that open-ended answers are simply *longer*, and long wrong answers might be trivially easy to flag (e.g. via degenerate or rambling generations), so the AUROC gain could be a length effect rather than a discreteness effect. It is not. Re-grading correctness with **token-F1** — a length-sensitive, partial-credit metric rather than the binary LLM-judge label — leaves the routing AUROC essentially unchanged (the two gradings agree), so the signal is not an artifact of answer length. The gain tracks the *discreteness of the target* (letter vs. free text), which is why even short free-text answers (SLAKE, VQA-RAD) show the lift. The open-ended discreteness claim is our own contribution, distinct from the cross-model agreement literature that motivates open-ended gates without isolating the MCQ-vs-open-ended mechanism.

5.4 Even open-ended, plain confidence is still the best gate

The ceiling-break makes routing work; it does *not* make a fancier training-free signal necessary. On the same open-ended slice (Lingshu-7B→32B, n = 845, predicting cheap-wrong), we ran a gate-hunt over the natural alternatives — self-consistency in exact and semantic forms, semantic entropy, mean token-F1 self-agreement, self-verification P(True), and a fusion of all signals — and none beats plain cheap-model confidence:

Gate signal (cheap-wrong AUROC, n = 845)	AUROC
Cheap-model confidence (margin)	0.866
Fusion (all signals)	0.866
Exact self-consistency	0.845
Mean token-F1 self-agreement	0.844
Semantic entropy	0.807
Semantic self-consistency	0.806
Self-verification P(True)	0.755

Confidence and the all-signal fusion tie at 0.866 — i.e. adding self-consistency, semantic-entropy, or self-verification to confidence buys nothing, and self-verification P(True) (0.755) is *worse* than plain confidence. So even in the favorable open-ended regime, *training-free* selection is still floored: the best you can do with a frozen model and no extra training is plain confidence, which is the same conclusion the luck floor reached, now shown to survive the format change. This is the negative that motivates the one positive lever left — a small amount of training.

5.5 Consequence

The open-ended ceiling-break reframes the pessimistic MCQ verdict: the "gate is saturated at AUROC $\approx$ 0.6" result is, in large part, a benchmark artifact of multiple-choice discreteness rather than a statement about medical VQA. Concretely, this changes how the rest of the paper is evaluated. Because (i) confidence-gated escalation is real open-ended (routing AUROC $\approx$ 0.87), (ii) recoverability nonetheless stays at the luck floor (pooled 0.591), and (iii) no training-free selection signal beats plain confidence even here, the productive question becomes whether a *trained* selector can escape the floor that a frozen model cannot. We therefore study selection and verification open-ended throughout — the setting of the trained outcome verifier (Method II), which is the one lever that does break it.

6. Method I — the Adaptive-Compute Cascade (ACC)

The luck floor (§4) says that over a *frozen* model no training-free rule improves the **gate** (which queries to escalate) or the **selection** (which sample to keep). ACC touches neither. It changes the third, under-examined decision — the **action**: *what* escalation actually runs. Instead of escalating straight from the cheap 7B to the most expensive 32B configuration, ACC cascades over **compute configurations of the same two models**. The observation it exploits is that the 32B's cost is almost entirely its *reasoning* trace — hundreds of decoded `<think>` tokens — yet on perception-style medical VQA that reasoning is *not* what fixes the cheap model's errors; raw *capacity* is. So ACC inserts the strong model's **fast, no-think** mode as an intermediate tier and reserves the slow think pass for the small residual of queries that genuinely need step-by-step reasoning. The result is an efficiency method (parity accuracy, far less compute), not an accuracy method; the accuracy lever is Method II (§8).

6.1 Three tiers over compute configurations

ACC is a three-tier cascade over compute configurations of the **same** cheap/strong model pair (batch-1 latencies annotated; "cap320" is the reduced `max_pixels` image budget and "fullres" the uncapped one, both defined in §3):

T0 = 7B no-think @ cap320	($\approx$ 0.21 s ; ~388 prefill tok, ~2 decode tok)
T1 = 32B no-think @ cap320	($\approx$ 0.34 s ; ~388 prefill tok, ~2 decode tok)
T2 = 32B think @ fullres	($\approx$ 11.34 s, batch-1 ; ~685 prefill tok, ~391 decode tok)

The two lower tiers answer *directly* (no `<think>` trace, ≈ 2 decode tokens each), so they are near-instant; only T2 pays for a long reasoning decode. A query flows $T0 \rightarrow T1 \rightarrow T2$, stopping at the first tier the gates accept. T0 handles the confident majority; T1 absorbs the escalations that need capacity but not reasoning; T2 fires only on the genuinely hard reasoning residual.

6.2 Why a no-think middle tier: reasoning over-thinks perception

The middle tier only makes sense if the 32B's *no-think* mode is competitive with — or better than — its *think* mode on the sets ACC targets. It is. On the four competent (perception) benchmarks the 32B is no worse, and often materially better, without thinking; the think mode's advantage appears only on the reasoning-heavy sets (MMMU, MedXpertQA), which ACC excludes from its headline scope (§11). Per-benchmark accuracy for the MedVLTinker pair (ALL-6 splits; exact-match):

benchmark	7B no-think	32B no-think	32B think
PMC-VQA	0.543	0.551	0.556
SLAKE	0.762	0.849	0.764
VQA-RAD	0.761	0.853	0.776
PathVQA	0.641	0.661	0.672
MMMU-medical	0.547	0.624	0.688
MedXpert-Reasoning	0.225	0.279	0.326
MedXpert-Understanding	0.256	0.292	0.385

The two perception sets where escalation actually helps show the strong model *thinking itself out of correct answers*: on **SLAKE the 32B scores 0.849 no-think vs 0.764 think (+0.085)** and on **VQA-RAD 0.853 vs 0.776 (+0.077)**. PMC-VQA and PathVQA are near-flat (0.551 vs 0.556; 0.661 vs 0.672), while think genuinely pays only where reasoning is the task (MMMU 0.688 vs 0.624; MedXpert-Understanding 0.385 vs 0.292). Pooled over the four competent sets, **32B-no-think 0.658 $\geq$ 32B-think 0.645**, and it reaches that at roughly **2 decode tokens vs ~ 477** for the think pass. The mechanism is symmetric over-thinking, not a free lunch: think *fixes* 11.5% of no-think's errors but *breaks* 11.1% of its correct answers, so on perception the two roughly cancel. (A per-question oracle that could pick the better *mode* would reach 0.683 on ALL-6 / 0.759 on COMPETENT-4, +10-11 points over either single mode — evidence that the mode choice carries real signal, even though, per the luck floor, that oracle is not achievable from any cheap training-free signal.)

Why the middle tier runs at cap320. Capping the strong-leg resolution makes T1 cheap without costing accuracy on perception. Across strong-leg configurations, pooled over the competent-4 (backbone% = fraction of the always-think compute):

strong-leg config	accuracy	backbone%
think @ fullres	0.6451	100%
no-think @ fullres	0.6582	66%
think @ cap320	0.6319	72%
no-think @ cap320	0.6463	39%

no-think @ cap320 recovers essentially all of the think @ fullres accuracy (0.6463 vs 0.6451) at **39%** of the compute — so it is the natural T1. At the per-benchmark level 32B-no-think@cap320 scores SLAKE 0.849, VQA-RAD 0.853, PMC-VQA 0.551, PathVQA 0.661, using ~ 375 prefill / ~ 2 decode tokens versus ~ 852 / ~ 477 for a think call ($\approx 28\%$ of a think call's cost).

6.3 The two gates

ACC has two decision points, both thresholding cheap confidence signals already defined in §3 (margin $m = \square_1 - \square_2$ over the greedy letter-candidate logprobs).

Tier-0 gate (margin). From the 7B no-think decode, output T0's answer iff its margin $m_0 \geq \tau_0$; otherwise escalate to T1.

Think gate (agreement, ACC-v2). Let $\hat{y}_{T0}$, $\hat{y}_{T1}$ be the two no-think predictions. Fire the expensive think tier only when the two no-think legs **disagree**, breaking ties by the 32B-no-think margin:

$$\text{disagree} = 1[\hat{y}_{T0} \neq \hat{y}_{T1}]; \quad s_1 = \text{disagree} + \varepsilon \cdot (-m_{T1}), \quad \varepsilon = 1e-6; \quad \text{fire T2 iff } s_1 > \tau_1 \quad (\tau_1 \approx 1). \quad (3)$$

The integer disagree term selects the disagreements; the tiny $\varepsilon \cdot (-m_{T1})$ term orders those disagreements by the 32B-no-think margin, so with $\tau_1 \approx 1$ the gate fires T2 only on the *lowest-margin* disagreements. Consequently think fires on **$\sim 15\%$** of ALL-6 queries, well below the **$\sim 32\%$** raw disagreement rate — the agreement signal is filtered down to the least-confident disagreements, where reasoning is most likely to help.

Variants. *ACC-v1* uses a plain margin think gate ($\text{fire_think} = 1[\mathbf{m}_{\{T1\}} < \tau_1']$); it sends more queries to think (think $\approx 19.5\%$ on ALL-6). *ACC-v2* is the agreement gate of Eq. (3) (think $\approx 15.1\%$). *ACC-v3* tightens the gate to the **conjunction** $\text{fire_think} = 1[\hat{y}_{\{T0\}} \neq \hat{y}_{\{T1\}}] \wedge 1[\mathbf{m}_{\{T1\}} < \tau_1']$, requiring both a no-think disagreement *and* a low 32B-no-think margin before paying for reasoning. The only fitted parameters are the scalar thresholds τ_0 , τ_1 (and τ_1' for v1/v3), chosen on **held-out PMC-VQA-train calibration** to reach always-32B-think parity at minimum latency; nothing else is learned. (*The agreement mechanism of Eq. (3) is the ABC cascade family; the contribution is the tiered compute-configuration it gates, not the gate itself — novelty is discussed in §11.*)

6.4 Cost accounting

For a cascade with per-tier cost c , fraction e_0 escalated past T0 and fraction e_1 reaching the think tier, the expected cost is Eq. (1):

$$C = c_{T0} + e_0 \cdot c_{T1} + e_1 \cdot c_{T2}. \quad (1)$$

The same equation is evaluated in three currencies — FLOPs, batch-1 latency, and NVML-integrated energy — with the per-tier costs measured at batch-1 (one model forward costs $F = 2 \cdot N \cdot (P+G)$ FLOPs, $N_7 = 7.6e9$, $N_{32} = 33.0e9$):

tier	config	prefill P	decode G	FLOPs ($\times 1e15$)	latency	energy
T0	7B no-think @ cap320	388	2	0.01	0.21 s	25 J
T1	32B no-think @ cap320	388	2	0.03	0.34 s	65 J
T2	32B think @ fullres	685	391	0.07	11.34 s (batch-1)	~ 6319 J

The two no-think tiers are *constant-energy* (≈ 25 J, ≈ 65 J) because they decode only ~ 2 tokens; the think tier's energy scales with its decode length ($E \approx 18.17 \cdot G - 107.5$ J, ≈ 254 W $\times$ 27.5 s at $G \approx 391$). (*Cost methodology note: an earlier co-resident $rt_cascade$ measurement put T2 at ≈ 26.6 s / 6994 J — think latency ≈ 0.072 s per decode token, fit $R^2 = 0.99$, the think leg $\approx 80\times$ the no-think leg. That co-resident number is superseded by the batch-1 native accounting; the canonical headline uses $T2 \approx 11.34$ s, from the always-32B-think measurement.*)

For ACC-v2 on ALL-6 the measured escalation rates are $e_0 = 71.7\%$ (past T0) and $e_1 = 15.1\%$ (to think). Plugging these into Eq. (1), the cost is dominated by the third term $e_1 \cdot c_{T2}$, because c_{T2} dwarfs both no-think tiers: the whole cascade cost is governed by *how often the think tier fires*. The saving is structural. A conventional 2-tier 7B $\rightarrow$ 32B-think cascade at the same parity sends $e_1 \approx 69\%$ of queries to the think leg; ACC's free no-think middle tier absorbs the capacity-limited escalations and cuts that to **15%**. Since the avoided think calls are $\sim$ two orders of magnitude more expensive than the tiers that replace them, they account for essentially the entire compute reduction. The resulting parity efficiency — the $-80\%/-95\%$ latency, FLOPs $\rightarrow$ 52%/25%, and $\sim 5\times/\sim 28\times$ energy figures, at guard 0 — is reported in the ACC results section (§7), together with the "it's the structure, not the gate" ablation that swaps Eq. (3) for every other training-free gate and the cross-family reproduction.

7. Results I — ACC: large efficiency gains at parity, and it is the structure not the gate

This section reports what the Adaptive-Compute Cascade (§6) buys, and — the more important point — *where the gain comes from*. The efficiency comes from the tiered **compute-configuration** (a cheap no-think middle tier that absorbs most escalations), not from a clever routing rule: hold the three-tier structure fixed and swap in every training-free gate from the literature and they all land on one accuracy-FLOPs-latency frontier. We then show the mechanism and the savings reproduce across five model families and three architectures — with the honest wrinkle that on families where reasoning does not help perception, ACC does the right thing by *not* escalating (graceful collapse to the cheap leg), and on one family (MedGemma) the win is only partial.

Protocol and metrics (recap). All numbers below are at **parity accuracy**, defined as the accuracy of the always-on strong reasoning model (always-32B-think). The two scalar gate thresholds τ_0 , τ_1 are frozen on held-out PMC-VQA-train calibration; every evaluation sample is held out. FLOPs% is the prefill-inclusive cascade cost divided by the always-32B-think cost (Eq. 1); **latency** and **energy** are **measured batch-1** per-tier costs (energy NVML-integrated). esc_0 is the fraction escalated past T0 to the no-think 32B tier; think is the fraction that reaches the expensive T2 think tier. guard is the number of benchmarks on which a system is *worse than the always-7B leg*, averaged over 20 held-out seeds (so it is fractional: guard = 0.05 means that in one seed out of twenty a single benchmark dipped below the 7B; guard = 0 is a clean never-worse-than-cheap guarantee).

7.1 ALL-6 at parity (MedVLThinker)

At parity on the full six-benchmark suite (MedVLThinker 7B/32B), with every training-free gate swapped into the *same* three-tier structure:

system	acc	esc ₀	think	FLOPs%	latency	energy	guard
always-7B-nt @cap320	0.5262	0%	0%	8.4%	0.13 s	19.9 J	0.00
always-32B-nt @cap320	0.5573	100%	0%	36.2%	0.23 s	77.8 J	0.00
always-32B-think [PARITY]	0.5723	100%	100%	100.0%	11.34 s	6318.8 J	0.00
Ours: ACC-v2 (agreement gate)	0.5693	71.7%	15.1%	52.0%	2.27 s	1181.9 J	0.00
ACC-v1 (margin gate)	0.5687	66%	19%	53.9%	2.69 s	1416.6 J	0.00
CASP-Stability (trained gate)	0.5698	74%	11%	49.0%	1.77 s	899.2 J	0.05
MSP/Chow	0.5697	70%	19%	57.4%	2.96 s	1568.1 J	0.00
entropy	0.5691	71%	21%	62.0%	3.48 s	1863.1 J	0.00
Gini/DOCTOR	0.5702	69%	22%	61.0%	3.44 s	1837.1 J	0.00
AutoMix (self-verify)	0.5692	73%	18%	54.6%	2.50 s	1307.0 J	0.05
FrugalGPT-style learned	0.5677	70%	19%	60.4%	3.30 s	1765.5 J	0.10
Jitkrittum L2D (Diff-Prob)	0.5666	67%	15%	50.6%	2.29 s	1194.5 J	0.00
random	0.5641	89%	76%	116.5%	8.95 s	4889.5 J	0.05

Headline (ALL-6): at -0.003 accuracy ($0.5723 \rightarrow 0.5693$, i.e. parity) and **guard 0** (never worse than the 7B on any benchmark), ACC-v2 takes **latency 11.34 s $\rightarrow$ 2.27 s (-80%)**, **FLOPs 100% $\rightarrow$ 52%**, and **energy 6318.8 J $\rightarrow$ 1181.9 J ($\sim 5.3\times$, -81%)**. The savings come entirely from the third tier: the fraction that reaches the expensive think pass drops from $\sim 69\%$ (what a naive 2-tier 7B $\rightarrow$ think cascade would spend) to **15.1%**, because the no-think 32B middle tier absorbs the bulk of escalations that need *capacity*, not *reasoning*.

Per-benchmark: guard 0 is real, and ACC wins exactly where think over-thinks. Breaking the ALL-6 accuracy out by benchmark shows why the aggregate parity holds and where the -0.003 comes from:

system	PMC	SLAKE	VQA-RAD	PathVQA	MMMU	MedXpert-R	MedXpert-U
always-7B-nt	0.543	0.762	0.761	0.641	0.547	0.225	0.256
always-32B-nt	0.551	0.849	0.853	0.661	0.624	0.279	0.292
always-32B-think [PARITY]	0.556	0.764	0.776	0.673	0.688	0.326	0.384
Ours (ACC-v2)	0.561	0.842	0.861	0.679	0.643	0.282	0.310

ACC-v2 is $\geq$ **the 7B on all seven benchmarks** (guard 0), and it actually *beats* the parity think baseline on the four perception sets — PMC (0.561 vs 0.556), SLAKE (0.842 vs 0.764), VQA-RAD (0.861 vs 0.776), PathVQA (0.679 vs 0.673) — precisely because 32B-no-think $\geq$ 32B-think on perception (the over-thinking premise, §6). It trails the think baseline only on the three *reasoning* sets (MMMU 0.643 vs 0.688, MedXpert-R 0.282 vs 0.326, MedXpert-U 0.310 vs 0.384), where real reasoning genuinely helps. The -0.003 aggregate deficit is therefore concentrated on MMMU/MedXpert; this is the honest scope boundary that makes ALL-5 (below) and COMPETENT-4 the settings where ACC is unambiguously at-or-above parity.

7.2 ALL-5 at parity (excluding near-chance MedXpert)

Removing MedXpertQA-MM — where both models are near chance and which is excluded from the ACC claims — sharpens the result, because the reasoning residual the think tier must service almost vanishes:

system	acc	esc ₀	think	FLOPs%	latency	energy	guard
always-7B-nt @cap320	0.6201	0%	0%	8.9%	0.13 s	19.9 J	0.00
always-32B-nt @cap320	0.6457	100%	0%	38.7%	0.23 s	77.8 J	0.00
always-32B-think [PARITY]	0.6463	100%	100%	100.0%	8.88 s	4915.9 J	0.00
Ours: ACC-v2 (agreement)	0.6450	35.1%	2.3%	24.9%	0.44 s	172.8 J	0.05
CASP-Stability (trained)	0.6461	43%	2%	28.3%	0.46 s	184.5 J	0.05

ACC-v1 (margin)	0.6435	32%	3%	24.8%	0.53 s	223.7 J	0.05
MSP/Chow	0.6444	39%	3%	27.0%	0.51 s	212.2 J	0.05
entropy	0.6459	50%	3%	31.2%	0.52 s	211.0 J	0.05
Gini/DOCTOR	0.6461	44%	2%	28.4%	0.46 s	179.4 J	0.10
AutoMix (self-verify)	0.6448	25%	2%	20.6%	0.35 s	129.7 J	0.65
FrugalGPT-style learned	0.6449	48%	3%	30.8%	0.56 s	235.2 J	0.20
Jitkrittum L2D (Diff-Prob)	0.6403	32%	2%	23.6%	0.41 s	161.3 J	0.20
random	0.6390	82%	17%	57.2%	1.79 s	899.7 J	0.40

Headline (ALL-5): always-think **8.88 s / 4915.9 J** → **ACC-v2 0.44 s / 172.8 J, FLOPs 24.9%** — a **−95% latency, ~28× energy** (−96%) reduction at parity (0.6463 → 0.6450). Note the honest caveat in this table: the *cheapest* FLOPs point is AutoMix at 20.6%, but it carries **guard 0.65** (it frequently drops below the 7B on some benchmark across seeds); ACC-v2 at **24.9% FLOPs / guard 0.05** is the honest operating point, and the trained CASP-Stability gate (28.3%, guard 0.05) is comparable. The learned gates (FrugalGPT-style guard 0.20, Jitkrittum L2D guard 0.20) are both cheaper on paper but less guardrail-safe — training the gate does not buy a cleaner frontier here (see §5).

7.3 It is the structure, not the gate

The two ALL-6 / ALL-5 tables above make the central claim visually: **holding the three-tier compute-configuration fixed and swapping the routing rule barely moves the operating point**. On ALL-6, every training-free gate — margin (ACC-v1), agreement (ACC-v2), MSP/Chow, entropy, Gini/DOCTOR, AutoMix self-verification, the FrugalGPT-style learned router, and the Jitkrittum L2D Diff-Prob deferral rule — lands in a narrow band: **accuracy 0.5666-0.5702, FLOPs 49.0-62.0%**, and **latency 1.77-3.48 s**, all at parity. The trained CASP-Stability gate (49.0% FLOPs, 1.77 s) and the agreement gate (52.0%, 2.27 s) are merely the cheapest points on that one frontier; agreement is the cheapest *guard-clean* one. Only **random** escalation leaves the frontier (116.5% FLOPs, 8.95 s) — i.e. the structure, not the decision rule, is what delivers the savings; a "better gate" moves you a few points along a frontier the structure already fixed. This is the empirical face of the luck floor (§5): recoverability is ~0.6 AUROC from any signal, so no gate can meaningfully out-route another.

Robustness. Bootstrap CIs on the ACC-v2 accuracy (from the held-out seed distribution) are **ALL-6 [0.5608, 0.5820]** and **ALL-5 [0.6372, 0.6562]**, both straddling the parity target; the batch-1 latency CI is wide (**[2.6, 9.8] s**), reflecting the heavy-tailed cost of the rare think calls at batch-1. ACC-v3 (a confidence-tightened conjunctive think gate) further trims think firings **19% → 14%** and FLOPs to **52.6%** at equal accuracy, holding parity on 20/20 seeds.

7.4 It generalizes across families and architectures

The "no-think ≥ think on perception" premise and the ACC savings were re-measured on four additional model pairs spanning three architectures (Lingshu-7B/32B, QoQ-Med-7B/32B, Chiron-o1-2B/8B, MedGemma-4B/27B), plus MedVLThinker for reference:

family	set	parity acc (32B-think)	ACC-v2 acc	ACC-v2 FLOPs%	esco	think	guard
MedVLThinker	ALL-6	0.5723	0.5693	52.0%	71.7%	15.1%	0.00
MedVLThinker	ALL-5	0.6463	0.6450	24.9%	35.1%	2.3%	0.05
Lingshu	ALL-6	0.6611	0.6614	48.6%	61%	0%	1.00
Lingshu	ALL-5	0.7746	0.7726	38.9%	42%	1%	1.15
QoQ-Med	ALL-6	0.4689	0.5095	8.8%	0%	0%	0.00
QoQ-Med	ALL-5	0.5432	0.6048	9.1%	0%	0%	0.00
Chiron-o1	ALL-6	0.5076	0.6023	19.3%	0%	0%	0.00
Chiron-o1	ALL-5	0.5926	0.7249	19.9%	0%	0%	0.00
MedGemma	ALL-6	0.5253	0.5219	68.4%	55%	20%	1.15
MedGemma	ALL-5	0.5979	0.6028	11.5%	0%	0%	0.00

Lingshu — a clean win. ACC matches or slightly beats parity while roughly *halving* FLOPs: **48.6%** on ALL-6 (0.6614 vs parity 0.6611) and **38.9%** on ALL-5 (0.7726 vs 0.7746). The guard of 1.00 (ALL-6) / 1.15 (ALL-5) is **inherited from the target, not caused by the cascade**: the always-on Lingshu-32B is itself worse than Lingshu-7B on one benchmark (MMMU, the known Lingshu-7B inflation anomaly, §11), so any system reaching for the 32B pays that guard. ACC-v1 (margin) is comparable (48.6% / 39.3%).

QoQ-Med and Chiron-o1 — graceful collapse. On these two families the strong model's *think* mode actively **hurts**: the cheap 7B/2B no-think leg already beats always-32B/8B-think (QoQ 0.5094 cheap vs 0.4689 think on ALL-6; Chiron

0.6024 vs 0.5076). ACC does the correct thing — it **does not escalate** ($\text{esc}_0 = 0\%$, $\text{think} = 0\%$), collapsing to the cheap leg and thereby delivering accuracy that *exceeds* the "parity" think target (QoQ 0.5095, Chiron 0.6023 on ALL-6; QoQ 0.6048, Chiron 0.7249 on ALL-5) at cheap-leg cost (**8.8% / 19.3% FLOPs**, guard 0). This is the intended failure mode: when reasoning is worthless, the agreement gate never wastes compute chasing it. It is a graceful degradation, not a savings on these families in the usual sense — there is simply nothing to escalate to.

MedGemma — a partial win. MedGemma is the honest mixed case. On ALL-6 the agreement gate escalates heavily ($\text{esc}_0 = 55\%$, $\text{think} = 20\%$) for only a modest **68.4% FLOPs** at guard 1.15, and the margin gate (ACC-v1) is worse still (93.8% FLOPs); here the trained CASP-Stability gate does noticeably better (29.6% FLOPs, guard 1.15). On ALL-5, MedGemma's cheap leg already beats big-think (0.6031 vs 0.5979), so ACC again collapses to the cheap leg (**11.5% FLOPs**, guard 0). So for MedGemma the efficiency win is real on ALL-5 but only partial / gate-dependent on ALL-6.

Takeaway. ACC delivers its large, guard-clean savings on the two families whose 32B has a genuine think-mode advantage that over-fires on perception (MedVLThinker, Lingshu); on families where think does not help it degrades gracefully to the cheap leg (QoQ, Chiron, and MedGemma-ALL-5); and the one partial case (MedGemma-ALL-6) is exactly where a trained gate helps and the agreement gate over-escalates. Across all of this the accuracy stays at parity and the guard stays clean or is inherited from the target — never manufactured by the cascade.

8. Method II — a Trained Outcome Verifier

The luck floor (§4) is a statement about *training-free* decision rules over a *frozen* generator: no confidence signal, self-verification, or aggregation rule beats trivial baselines, because the correct answer is a *minority* candidate in most recoverable questions (the majority trap) and the strong model fails on the same inputs the cheap one does. This section supplies the constructive complement — a small **trained outcome verifier** — and defines it precisely; §9 reports the accuracy it buys (across three model families, held-out OOD, and structured bounding boxes). The key claim of this section is *methodological*: the verifier is a supervised, inference-time **selector**, not a new gate and not a smarter generator, and it is trained only on labels a dataset answer key already contains — which is why it escapes the luck floor without importing any capability from the model it is later compared against.

8.1 Why a trained verifier, and why here specifically

In *general* LLM reasoning, a trained-verifier best-of-N barely beats plain self-consistency: recent work reports near-parity between learned verifiers and simple confidence/aggregation baselines (self-certainty best-of-N, NeurIPS 2025 [2502.18581]; aggregation studies [2510.13918]). Under that evidence a learned selector is often judged not worth its supervision cost.

Medical open-ended VQA is the *opposite* regime, and it is the regime in which a learned selector should pay off. Three conditions hold simultaneously, each documented earlier:

1. **Self-consistency fails.** The correct answer is a minority vote in 74–90% of recoverable questions (the majority trap, §4), so any rule that rewards agreement — SC-majority, weighted voting, self-verification — sits at or below the random-pick floor.
2. **The strong model barely helps.** On open-ended answers the $5\times$ model recovers only a small, near-unlearnable slice of the cheap model's errors (the recoverability wall, §4), so "just scale up" is not a fix either.
3. **The oracle headroom is large.** Over $N=8$ samples from the *cheap* generator, a perfect selector reaches far above greedy (pooled oracle@8 0.592 vs greedy 0.413, §9). The right answer is usually *present in the sample set*; what is missing is a rule that *picks* it.

A selector is exactly the tool for (3) when (1) and (2) rule out the cheap and the expensive shortcuts. This is why a *learned* verifier — rather than a better gate or a bigger model — is the correct instrument in this setting.

8.2 The verifier and best-of-N selection

The verifier is a LoRA-adapted VLM (base **Lingshu-7B** for free-text answers; **Qwen2.5-VL-7B** for boxes) with adapter parameters ϕ . Given the image v , question q , and a candidate answer a , it scores the candidate by the probability it assigns to the token "Yes" versus "No" at the final position of the prompt "*...Is the proposed answer correct? Answer Yes or No.*":

$$s_\phi(v, q, a) = P_\phi(\text{Yes} \mid v, q, a) = e^{\{z_{\text{Yes}}\}} / (e^{\{z_{\text{Yes}}\}} + e^{\{z_{\text{No}}\}}). \quad (4)$$

$s_\phi \in [0, 1]$ is a per-candidate correctness score. The base model is frozen; only the LoRA adapter (≈ 190 MB) is trained, on per-sample correctness labels y (defined in §8.3) by binary cross-entropy on the Yes/No token:

$$L(\phi) = - \sum [y \cdot \log s_\phi + (1 - y) \cdot \log(1 - s_\phi)]. \quad (5)$$

Given N candidates $a_1 \dots a_N \sim M(\cdot \mid v, q)$ drawn from the *same cheap generator* M , **best-of-N selection** returns the highest-scoring candidate:

$$\hat{a} = \operatorname{argmax}_{\{i \leq N\}} s_{\varnothing}(v, q, a_i).$$

(Eq. 4 applied N times)

We report the quality of $\hat{a}$ with the gap-captured metric of Eq. (2) (§3), $(\operatorname{acc}(\hat{a}) - \operatorname{greedy}) / (\operatorname{oracle@N} - \operatorname{greedy})$, i.e. the fraction of the achievable oracle headroom that the selector realizes; and, in the cascade of §9, the same $s_{\varnothing}$ doubles as an escalation signal (escalate iff the selected candidate's score is low), so one trained head serves both selection and gating. This is a **pointwise outcome verifier** — it scores a *completed* answer, not intermediate reasoning steps — trained for inference-time selection; its lineage and how it differs from process-reward and generative-verifier work is discussed in §2. argmax over $s_{\varnothing}$ is the intended rule (voting-style aggregation reintroduces the majority trap; §9 quantifies the cost of the alternatives). Training is deliberately small: $\approx 6,000$ judged (candidate, label) pairs at a positive rate of 0.194, one LoRA adapter, no change to the generator or the strong model.

8.3 Label rules: free-text answers and bounding boxes

The only thing the verifier is supervised on is the label y — whether a candidate is correct. We use the **same** machinery (Eq. 4–5, argmax selection) for two output types, differing only in how y is defined.

Free-text answers. y is produced by an LLM judge that decides whether the free-text candidate a *matches the gold answer* — a semantic stand-in for exact-match, which is too brittle for open-ended text (synonyms, bilingual answers, paraphrase). The judge is **MedVLThinker-32B**, cross-checked by an independent **Claude-Sonnet-5** judge; its validation (100% agreement on exact-match anchors; second-judge κ 0.85–0.96) is reported in §4. Crucially, the judge compares the candidate to the dataset **answer key**, so y encodes match-to-gold, not the judge's own medical opinion (see §8.4).

Bounding boxes. For grounding, a is a predicted box and the label is a hard IoU threshold against the gold box:

$$y = 1[\operatorname{IoU}(a, \text{gold}) \geq \theta], \quad \theta = 0.3.$$

(label rule, boxes)

The box-verifier (base Qwen2.5-VL-7B) is otherwise identical — Eq. (4)–(5), best-of-8 selection — and judges "does this drawn box localize the {target}?" One implementation detail matters: raw Qwen box coordinates are mis-scaled on large images (unscaled candidates give mean IoU only 0.098, $\operatorname{oracle@0.3} = 0.253$), so candidate and gold boxes are rescaled by the image width/height before computing IoU; with that fix the same selection recipe applies. Results on SLAKE organ boxes and the real MS-CXR chest-X-ray benchmark are in §9.

8.4 Why a 32B-judged 7B verifier beating the 32B is not circular

The results in §9 show a 7B-based verifier that matches — and on hard sets modestly beats — a $5\times$ strong model, using labels produced by a 32B judge. This invites a circularity objection ("the 32B is grading its own superior"), which the construction rules out on four grounds.

1. **The judge is an automated grader, not a knowledge oracle.** Its sole function is to decide whether a candidate string matches the **gold** answer. The labels y therefore derive from the dataset **answer key**; the grader could be a human or exact-match, and is a judge only because free-text needs a semantic comparator. No medical knowledge of the 32B *generator* enters the labels.
2. **The verifier is trained only on the cheap model's own samples.** It never sees a 32B answer; it learns to *discriminate* among the 7B's candidates. Discrimination is strictly easier than generation: the cheap model's 8 samples already contain a judged-correct answer in $\approx 59\%$ of pooled cases ($\operatorname{oracle@8} = 0.592$, §9), so the verifier need only *identify* the correct candidate that is already present — which is exactly why a **7B** selector suffices and imports no 32B capability.
3. **The comparison is between two deployment strategies, not two knowledge levels.** One strategy is scaling to a $5\times$ model *zero-shot*; the other is sampling a small model N times and selecting with a small *supervised* verifier. The in-domain supervision (a few thousand gold labels) is precisely the contribution — and its cost — not a leak of strong-model knowledge.
4. **The verifier genuinely uses the image.** Across all 8,512 candidates it separates correct from incorrect at **AUROC 0.924** (mean score 0.749 for correct vs 0.171 for incorrect), and a blank-image ablation drops discrimination by **−0.047** (VERIFIED_FACTS §D) — it degrades when the image is removed, so it is doing visual verification rather than mimicking a text-only judge (the judge, unlike the verifier, needs the gold answer). This directly refutes the "lazy verifier" failure mode of Verification-Mirage [2605.10850].

Because zero-shot self-verification is luck-floored (self-verification AUROC ≈ 0.5 , §4) while the *trained* head reaches AUROC 0.924, the discriminative signal is present but not zero-shot-surfaceable: **training is the active ingredient**, and it is small, in-domain, and label-only. The honest scope of the resulting win — a tie-to-modest gain versus the strong model, two-seed-robust over training-free baselines, with a wide per-dataset spread and a weak-base failure — is quantified in §9.

9. Results II — the verifier beats the 32B across three families

The trained outcome verifier of §8 is used exactly as defined there — best-of-8 selection over a small model's own samples, $\hat{a} = \operatorname{argmax}_i s_\phi(v, q, a_i)$ (Eq. 4), plus a verifier-confidence escalation gate that hands the lowest-confidence picks to the strong model. This section reports what that buys. The headline is a *positive*, and a strong one: across **three model families and two model architectures**, the 7B-verifier cascade **beats the strong 32B/38B model** on accuracy on every family×dataset cell we measured, including a held-out out-of-distribution set the verifier never saw. We then subject the win to the strictest possible test — a single held-out same-split comparison across two random seeds — where it narrows honestly to a *modest-win-to-tie*, and we characterize the ceiling that stops it from going further (candidate quality, not the selector).

9.1 The three-family beats-32B master table

The verifier base is **Lingshu-7B** throughout; for the InternVL3 family we apply the *same* Lingshu-trained verifier to InternVL3's answers (no retraining) — a **cross-architecture transfer**. Each family is evaluated on its full open-ended sets. "cheap(SC)" is the small model's self-consistency majority; "STRONG" is the family's 32B/38B single pass; "verifier-bo8" is best-of-8 verifier selection with no gate; "cascade-best@esc" is verifier-bo8 *plus* the verifier-confidence gate, escalating the stated fraction to the strong model; "oracle@8" is the luck ceiling (an oracle that always picks the correct sample when one exists among the 8).

Lingshu family (strong = Lingshu-32B):

dataset	cheap (SC)	STRONG (32B)	verifier-bo8	cascade-best @esc	oracle@8
VQA-RAD	0.465	0.600	0.575	0.625 @24%	0.630
PathVQA	0.324	0.376	0.453	0.469 @33%	0.517
Kvasir	0.286	0.301	0.439	0.448 @20%	0.491
RadImageNet-OOD	0.329	0.289	0.353	0.353 @0%	0.512
POOLED	0.322	0.331	0.414	0.421 @12%	0.513

MedVLThinker family (strong = MVT-32B):

dataset	cheap (SC)	STRONG (32B)	verifier-bo8	cascade-best @esc	oracle@8
VQA-RAD	0.420	0.525	0.490	0.555 @54%	0.600
Kvasir	0.343	0.361	0.477	0.483 @9%	0.550
RadImageNet-OOD	0.204	0.202	0.241	0.243 @13%	0.317
POOLED	0.266	0.277	0.339	0.344 @14%	0.416

InternVL3 family (strong = IV3-38B; Lingshu-trained verifier — cross-architecture transfer):

dataset	cheap (SC)	STRONG (38B)	verifier-bo8	cascade-best @esc	oracle@8
VQA-RAD	0.445	0.415	0.570	0.580 @12%	0.620
PathVQA	0.081	0.096	0.116	0.125 @52%	0.192
Kvasir	0.362	0.380	0.479	0.487 @17%	0.593
RadImageNet-OOD	0.285	0.304	0.302	0.313 @52%	0.398
POOLED	0.202	0.218	0.249	0.255 @36%	0.337

Reading the table (with the caveats stated). The **cascade-best** column beats **STRONG** on *every* cell of *every* family — pooled, that is Lingshu **0.421 vs 0.331**, MedVLThinker **0.344 vs 0.277**, InternVL3 **0.255 vs 0.218**. Two honest qualifications sit inside this. (i) **The gate is doing real work where the strong model is genuinely better.** Pure verifier-bo8 (no gate) does *not* uniformly beat the strong model: it loses on Lingshu VQA-RAD ($0.575 < 0.600$), MedVLThinker VQA-RAD ($0.490 < 0.525$), and InternVL3 RadImageNet ($0.302 < 0.304$). On exactly those sets the verifier-confidence gate escalates a larger fraction (24%, 54%, 52%) and the cascade climbs *above* the strong model by combining best-of-N on the recoverable questions with deference on the rest — the cascade wins by *knowing when to escalate*, not by out-selecting the 32B everywhere. (ii) The **POOLED** strong numbers here ($0.331 / 0.277 / 0.218$) are low because these full-set pools are dominated by hard and OOD sets (Kvasir, RadImageNet) and exclude SLAKE, the one set where the 32B is dominant; the stricter same-split test in §9.2 puts the 32B back at a higher 0.462, so we do not lean the whole claim on this pooled figure. (iii) On InternVL3 PathVQA the absolute numbers are near the floor for the whole family ($0.081/0.096/0.125$) — the *relative* cascade win holds, but nobody should read 0.125 as a usable operating point. The most defensible one-line summary: **the full-set 3-family cascade beats the strong model on every cell, and the Lingshu verifier transfers cross-architecture to lift a 38B InternVL3 family it was never trained on.**

9.2 A rigorous same-split test, and honest two-seed accounting

The full-set tables above pool over whatever sets each family completed. For a clean apples-to-apples comparison we also report a single **canonical held-out same-split** test: pool the four Lingshu open-ended sets, hold out 30%, and evaluate every method — including the 32B — on the *identical* held-out questions ($n=1064$).

dataset	greedy	self-consist.	32B (same split)	verifier	oracle	n
SLAKE	0.738	0.738	0.829	0.762	0.895	210
VQA-RAD	0.519	0.500	0.648	0.611	0.722	54
PathVQA	0.352	0.349	0.377	0.441	0.513	435
Kvasir	0.282	0.282	0.326	0.405	0.493	365
POOLED	0.413	0.411	0.462	0.501	0.592	1064

The ordering is **verifier 0.501 > 32B 0.462 > SC 0.411 $\approx$ greedy 0.413**: the trained 7B verifier captures **49%** of the oracle gap and, pooled, edges past the 32B *while being 5 $\times$ smaller*. The per-dataset pattern is the honest one: the verifier **beats the 32B on the hard sets** (PathVQA 0.441 vs 0.377, Kvasir 0.405 vs 0.326) where scaling to a bigger model stalls, and **loses to the 32B where the 32B is genuinely stronger** (SLAKE 0.762 vs 0.829) or where n is tiny (VQA-RAD 0.611 vs 0.648, n=54).

Two-seed honesty. On the seed-0 split the paired bootstrap gain of the verifier over the 32B is **+0.0385, 95% CI [+0.010, +0.066] — excludes 0** (significant). On a second 70/30 split (seed-1) it is **−0.005 (a tie: 0.445 vs 0.450)**. So the careful claim versus the 32B is **matches / competitive, mean +0.017 — a modest win to a tie**, not a robust rout. What is robust across both seeds is the margin over training-free selection: the verifier beats greedy by **+0.088 (seed-0) / +0.066 (seed-1)** while self-consistency $\approx$ greedy (paired bootstrap vs first-sample **+0.116, 95% CI [+0.092, +0.140]**), and gap-captured is **35–49% (mean $\sim$ 42%)**, driven by the hard sets — it helps PathVQA and Kvasir on *both* seeds and is flat where the baseline is already high (SLAKE, seed-1 −4% at greedy 0.774) or n is tiny (VQA-RAD, $\sim$ 0% at n=69). This is the sense in which "beats the 32B across three families" (§9.1, full-set cascade) and "matches-to-modest-win versus the 32B" (this strict same-split) are both true and must be reported together.

9.3 Training is the active ingredient

The gain comes from the *supervised* verifier, not from best-of-N or from the base model's own confidence. **Zero-shot self-verification** — asking the same model P(True) about its candidates — is luck-floored (§5). Training the same 7B on judge labels lifts it to 0.501; and, decisively, the **trained 7B verifier beats a zero-shot 32B verifier: 0.403 vs 0.355** selection accuracy on a matched 600-question sample (oracle@8 0.490), with selection efficiency **0.810 (7B-trained) vs 0.717 (32B-zero-shot)** — i.e. task-specific training beats 5 $\times$ the parameters for the selection job. The verifier's scores separate correct from incorrect candidates at **AUROC 0.924** across all 8,512 candidates (mean score 0.749 vs 0.171), and a blank-image ablation drops discrimination **−0.047 (VERIFIED_FACTS §D)** — it is reading the image, not mimicking the judge (which needs the gold answer; the verifier does not). (Fig. [figs/limits/fig_verifier_discrimination.png](#).)

Argmax is the correct selection rule. Given the verifier's scores, the plain argmax over candidates (0.501) beats **verifier-weighted voting (0.489, −0.012 [−0.024, 0.000])** and a **score $\times$ count hybrid (0.470, −0.031)**, and both beat plain majority (0.411). Any vote-flavored rule reinherits the majority trap (§5): the correct answer is a minority vote in most recoverable questions, so mixing counts back in only hurts.

9.4 Generalization: held-out sets, other generators, other bases

The verifier is not memorizing one dataset's answer distribution.

- **Held-out OOD (zero-shot).** The pooled-4 verifier applied *unchanged* to a fifth set, RadImageNet-VQA, lifts greedy **0.329 $\rightarrow$ 0.353** (13.6% of its gap, oracle 0.512, n=2000) — and that 0.353 already exceeds the Lingshu-32B's 0.289 on the same set. On Kvasir (a GI-endoscopy OOD set) the two-dataset verifier lifts **0.2858 $\rightarrow$ 0.3267** (19.9%, oracle 0.4908, n=1200).
- **Cross-generator (fixed verifier, different answers).** Applied to a *different model's* answers — the Lingshu-trained verifier scoring MedVLThinker-7B candidates — it still captures **49% (SLAKE) / 61% (VQA-RAD)** of the oracle gap; the verifier is largely generator-agnostic. (The InternVL3 column of §9.1 is the same story at the family scale.)
- **Other base, from scratch (honest, mixed).** Trained *from scratch* on a MedVLThinker-7B base it works on SLAKE (**0.564 $\rightarrow$ 0.622, 42%**) and is pooled-positive (**0.547 $\rightarrow$ 0.583, 25%**), but *fails* on VQA-RAD's tiny split (**0.500 $\rightarrow$ 0.470, n=54**). So the method is **not uniformly robust across bases**, and a stronger base makes a stronger verifier (the Lingshu verifier's 35–49% / cross-generator 49–61% exceeds this from-scratch MedVLThinker verifier's 25% pooled). The validated headline is the Lingshu result (four datasets, two seeds); the from-scratch other-base result is reported as an honest mixed finding, not a headline.

9.5 A test-time-scaling method: compute rivals parameters where parameters don't help

Best-of-K accuracy rises monotonically with the sample budget while a random pick stays flat, the defining test-time-scaling property:

K	1	2	4	8
verifier best-of-K	0.385	0.425	0.476	0.501

FLOPs (7B-fwd-eq)	1×	4×	8×	16×
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(oracle@8 = 0.592; random pick $\approx$ 0.39, flat; Fig. [figs/limits/fig_verifier_scaling.png](#).) Extending to K=16 on a fresh, larger pool it keeps rising with diminishing returns (0.356 $\rightarrow$ 0.394 $\rightarrow$ 0.411 $\rightarrow$ 0.417 $\rightarrow$ 0.424). Because reasoning barely helps open-ended medical VQA (§6), the strong model's single pass is a weak baseline to beat: pooled same-split, the **32B scores 0.462**, *matched-to-beaten* by the 7B with verifier-bo8 (**0.501**, seed-0; a tie 0.445 vs 0.450 on seed-1). The compute premium is real and stated honestly — best-of-8 costs **16× a 7B forward** versus the 32B's **4.57× a 7B forward** ($\approx 3.8\times$ the FLOPs; sheet:faithful §5) — but it is not all cost: because $2N < 4.57$ for $N \leq 2$, the **verifier-bo2 with no gate beats the 32B on accuracy and FLOPs (4.0 vs 4.57 7B-fwd-eq)** on the sets where the strong model is weak/OOD, so there is a genuinely dominating operating point on hard data. The verifier is thus the accuracy-optimal operating point on the hard sets, not dominated by simply renting a 5×-larger model. (Fig. [figs/limits/fig_verifier_pareto.png](#).)

9.6 Structured outputs: the win holds for bounding boxes

The same recipe — an IoU-thresholded label $y = 1[\text{IoU}(a, \text{gold}) \geq 0.3]$, a LoRA box-verifier (Qwen2.5-VL-7B base), best-of-8 selection — transfers to grounding, on organ boxes and on a **real** PhysioNet chest-X-ray benchmark.

benchmark	greedy	SC-medoid	zero-shot box-verifier	trained box-verifier	oracle@8	gap captured
SLAKE organs (n=487)	0.197	0.164	0.177	0.255 / 0.257	0.343	40% / 53%
MS-CXR pathology (n=435, real)	0.041	0.053	0.115	0.232 / 0.230	0.285	78% / 77%

Both training-free selectors sit at the luck floor — SC-medoid (0.164) is *below* greedy (0.197) on SLAKE — while the **trained** box-verifier recovers **40%/53%** (SLAKE, two seeds) and **78%/77%** (MS-CXR, two seeds) of the oracle gap. On MS-CXR the gain over greedy is **+0.191, 95% bootstrap CI [+0.152, +0.232]** (n=435), a **5.6× lift** — and training is again the active ingredient: the *zero-shot* box-verifier reaches only 0.115 (30% of the gap), so training roughly doubles the captured gap. The honest scope is that this lifts *selection over a weak base grounder*, not a trained SOTA grounder in absolute IoU; the point is that the "train a verifier to break the luck floor" principle extends from free-text answers to structured outputs. (Fig. [figs/limits/fig_trained_verifier_unified.png](#).)

9.7 The escalation gate: verifier-confidence is the best signal, and no trained gate beats it

Given best-of-N selection, *which* picks should we escalate to the strong model? The answer mirrors the luck floor of §5 exactly: **the verifier's own confidence is the best available gate, and no trained gate beats it by more than noise**. Predicting pick-correctness pooled, verifier-confidence reaches AUROC **0.853 / 0.885 / 0.875** (Lingshu / MedVLThinker / InternVL3), and the best trained gate (a GBM/logit on verifier-plus-cheap features) reaches **0.861 / 0.882 / 0.879** — a Δ of **+0.008 / -0.003 / +0.004**, i.e. noise, with identical cascade accuracy. A controlled gate-swap (verifier-bo8 fixed at 0.414, strong 0.331; Lingshu pooled) ranks the signals by area-under-the-deferral-curve (ADC, higher is better):

escalation gate	AUROC (pick-correct)	ADC
verifier-confidence [deployed]	0.853	0.3923 (best)
trained GBM (verifier + cheap)	0.854	0.3914 (ties, -0.001)
trained logit (verifier + cheap)	0.853	0.3911 (ties)
verifier-mean	0.834	0.3855
SOTA Diff-Prob (Jitkrittum L2D)	0.708	0.3832
self-consistency / -n_distinct	$\sim$ 0.69	$\sim$ 0.368
verifier-margin / verifier-negstd	0.40 / 0.44	$\sim$ 0.36

Cheap-only gates (self-consistency, n_distinct, cheap-confidence) land at **0.66-0.79**, far below; the recoverability-ranking gate in the Jitkrittum L2D style nudges its own AUROC 0.60 $\rightarrow$ 0.63-0.65 but yields no better cascade accuracy, and its ADC (0.3832) sits below verifier-confidence. The reason is the same **recoverability wall** we hit for routing between frozen models: the strong model fixes only **6-10%** of the verifier's errors (up to **26%** on the sets where the strong model is competitive), and *which* ones is near-unlearnable (AUROC $\sim$ 0.4-0.6; on InternVL3 pooled the recoverability signal is *below* chance, 0.367).

The deployed operating point. Adding the verifier-confidence gate on top of best-of-N selection, on the same n=1064 held-out split as §9.2, gives the accuracy-optimal **verifier-augmented cascade: 0.517 @ 35% escalation** — above verifier-alone (0.501) and the 32B (0.462), at a compute premium (≈ 17.5 7B-forward-equivalents). Per dataset (seed-0) it beats the 32B on Kvasir (0.422 @36%), PathVQA (0.453 @30%) and VQA-RAD (0.667 @37%), and only *matches* it on SLAKE (0.829 @100% — the verifier alone, 0.762, is below the 32B there, so the gate simply escalates everything). This is the same honest picture as the full-set tables: the cascade wins by out-selecting the strong model on the hard sets

and by deferring to it on the one set where it dominates.

9.8 The binding limit is candidate quality, not the selector

The verifier does not capture the whole oracle gap, and the diagnostics say precisely why: **the ceiling is the quality of the 8 candidates, not the selector's ability to rank them**. Per-answer, the verifier already ranks correct above incorrect at AUROC ~ 0.90 in every family (Lingshu 0.903, MVT 0.913, IV3 0.898), yet its **selection efficiency** — how often argmax picks a correct candidate *when a recoverable one exists* — is only **81% (Lingshu) / 82% (MVT) / 74% (IV3)**. Pushing per-answer ranking harder does not help selection: adding a within-question Bradley-Terry ranking loss raises per-answer AUROC **0.903 $\rightarrow$ 0.931 / 0.933** but leaves selection accuracy **flat at ~ 0.50** . What *does* move the ceiling is better candidates: pooling samples across the three model families raises **oracle@8 by +0.11-0.15** (VQA-RAD +0.150, Kvasir +0.138, RadImageNet +0.113), and the oracle already dwarfs the verifier in-family (oracle@8 vs verifier-bo8: Lingshu 0.513 vs 0.414, MVT 0.416 vs 0.339, IV3 0.337 vs 0.249). So the remaining gap is *headroom in the candidate set*, not a smarter loss — which is exactly why cross-model candidate pooling, not a better verifier, is the natural next lever (§12).

10. Results III — the faithful cross-family MCQ cascade and a reasoning think tier

§4 (the ACC) established the structural claim — *spend compute on the query, not on every query* — on the internal harness, with a single MedVLThinker family, and with the mechanism running the "wrong" way for reasoning (no-think beats think on perception). This section stress-tests that claim in the two directions where a systems contribution most needs to survive: **(a) a faithful, third-party reproduction protocol** (MedEvalKit, so the accuracies are directly comparable to published papers, not to our own harness), and **(b) three architecturally distinct model families** (Lingshu, MedVLThinker, InternVL3). It also adds the mirror-image structural lever the ACC omits: a **reasoning "think" tier** that *adds* accuracy on the reasoning benchmarks where thinking helps rather than hurts. Two structural moves, both regime-adaptive; neither is a "better gate."

Two honesty notes bound this section up front. First, the protocol here is the **faithful MedEvalKit pipeline** defined in §4 (isolated medeval_venv + the Qwen2_5_VL wrapper + native weights + use_vllm + TORCHDYNAMO_DISABLE, vLLM 0.9.0.1, MCQ graded exact-match with use_llm_judge=False); the internal NGC harness is **not** faithful and its numbers are not used here. Second, several July cells are still **provisional** — the InternVL3-38B MedXpert cell (a token-cap failure), the 7th benchmark (OmniMedVQA, still running), and all InternVL3/PathVQA *latency* figures (measured under GPU contention) — and are flagged as such below rather than folded into the headline.

10.1 Reproduction fidelity: the pipeline reproduces published baselines

Before trusting any cross-family cascade number, we confirm the faithful pipeline reproduces the published Lingshu baselines. Our MedEvalKit accuracies vs the Lingshu-paper (arXiv 2506.07044) figures, in percent:

benchmark	Lingshu-7B (ours)	7B (paper)	Lingshu-32B (ours)	32B (paper)
MMMU-Med	80.0	54.0	63.3	62.3
SLAKE-closed	82.5	83.1	85.9	89.2
PMC-VQA	54.3	56.3	55.2	57.9
MedXpert-MM	26.2	26.7	30.6	25.7
VQA-RAD-closed	78.1	67.9	85.3	76.5

The clean like-for-like anchor is **MMMU-Med-32B: ours 63.3 vs paper 62.3 — an exact match** ($0.633 = 62.3$). PMC-VQA is 4-option MCQ throughout and tracks the paper closely for both sizes (55.2 vs 57.9; 54.3 vs 56.3). The SLAKE and VQA-RAD rows are *not* strictly like-for-like — the paper reports the **combined** open+closed set (83.1 / 89.2 and 67.9 / 76.5), whereas our column is the **closed subset** that the MCQ cascade actually runs on — so their offsets (VQA-RAD-closed higher, SLAKE-closed lower) are composition, not error; on MedXpert-MM our reproduction even slightly exceeds the paper (30.6 vs 25.7). The one genuine anomaly is **Lingshu-7B MMMU-Med at 80.0**, a Lingshu-7B-specific inflation of roughly +26 over the paper's 54.0; it is a known quirk of that checkpoint, **excluded from all claims**, and confirmed to be family-specific because **MedVLThinker-7B on the identical eval is a normal 0.533** ($<$ its own 32B). As an independent cross-check we also reproduced the *open-ended* halves through a Claude-Sonnet-5 judge: **VQA-RAD-32B 74.1% (paper 76.5), SLAKE-32B 85.0% (paper 89.2)**, with the judge validated at a **100% exact-match anchor** (zero-word-overlap "correct" cases were all legitimate synonyms or bilingual Chinese matches) and independent-2nd-judge agreement κ **0.85-0.96**. The faithful pipeline is trustworthy; the cascade numbers below are built on it.

10.2 The 2-tier margin cascade matches the strong model at $-17\ldots-69\%$ FLOPs, across three families

The construction is deliberately the simplest possible instance of the ACC structure: a **2-tier cascade** — cheap 7B/8B tier, escalate to the strong 32B/38B tier — gated by the **same confidence-margin rule** used throughout the paper (margin $m = \square_1 - \square_2$; escalate iff $m < \tau$), with τ set to the iso-strong operating point (just enough escalation to match the

strong model's accuracy). Cost is the 2-tier form of Eq. 1, $C = 1 + \text{esc} \cdot 4.57$ in 7B-forward-equivalents (the cheap tier always runs at cost 1; the strong tier costs 4.57 and runs on the escalated fraction), and FLOPs-savings vs always-strong is $1 - C/4.57$.

The 3-family $\times$ 7-benchmark matrix of FLOPs savings at iso-strong accuracy:

benchmark	Lingshu	MedVLThinker	InternVL3
MMMU-Med	keep-cheap (7B anomaly)	−14%	−62%
PMC-VQA (33k)	−69%	−49%	−16%
SLAKE	−56%	no win (7B weak)	keep-cheap (8B $\geq$ 38B)
VQA-RAD	−17%	−41%	−67%
PathVQA	−31%	−68%	−20%
MedXpert-MM	no win (floor)	no win	(cap gap, fixup)

Read as a whole, the cascade **matches the strong model at −17...−69% FLOPs** wherever it wins, and the pattern of *where* it wins is governed by a single rule: **the win magnitude tracks the (strong − cheap) accuracy gap**. A *small* gap means the cheap model already answers most queries, so only a thin residual escalates and the saving is large; a *large* gap (the cheap model is weak) forces the gate to escalate nearly everything, and the extra cheap pass on top of an almost-always-strong cascade makes it *more* expensive than always-strong. This produces two non-winning regimes, each for a principled reason:

- **keep-cheap** (the small model already $\geq$ the big one, so escalation buys nothing): **InternVL3-SLAKE** (the 8B matches or exceeds the 38B) and **Lingshu-MMMU** (the inflated Lingshu-7B, §10.1).
- **no win** (the small model is too weak, or both are at the floor): **MedVLThinker-SLAKE** (7B 0.498 vs 32B 0.620 — a 12-point gap forces **96%** escalation and **+18%** FLOPs), and every **MedXpert-MM** cell (both models near the 4-option floor, $\sim$ 100% escalation, +17...+22% FLOPs). The **InternVL3-MedXpert** cell is not a floor result but an *unmeasured* one — see §10.5.

10.3 Per-benchmark cost, efficiency summaries, and threshold honesty

Per-benchmark detail for Lingshu (accuracy is exact-match; 7B = 1, 32B = 4.57 forward-equivalents):

benchmark	7B	32B	2-tier	esc%	Δ FLOPs	Δ latency
PMC-VQA (n=33,430)	0.543	0.552	0.552	9%	−69%	−33%
SLAKE-closed (836)	0.825	0.859	0.861	22%	−56%	−22%
VQA-RAD-closed (251)	0.781	0.853	0.853	61%	−17%	+21%
MedXpert-MM (2000)	0.262	0.306	0.307	95%	+17%	+60%
MMMU (150)	0.80*	0.64	—	—	—	—

(*Lingshu-7B MMMU inflated, §10.1.) And for MedVLThinker:

benchmark	7B	32B	2-tier	esc%	Δ FLOPs
PMC-VQA (33k)	0.521	0.537	0.537	29%	−49%
VQA-RAD-closed (251)	0.765	0.865	0.865	37%	−41%
MMMU (150)	0.533	0.613	0.613	64%	−14%
SLAKE-closed (836)	0.498	0.620	—	96%	+18% (no win)
MedXpert (2000)	0.239	0.299	—	100%	+22% (no win)

MedVLThinker's **PathVQA** cell is the family's largest saving (−68%, matrix above); its per-benchmark 7B/32B/escalation breakdown is not yet in these tables. For **InternVL3** we currently have only the matrix cells (MMMU-Med −62%, VQA-RAD −67%, PMC-VQA −16%, PathVQA −20%; SLAKE keep-cheap), with the MedXpert cell pending (§10.5).

Latency caveat. The Lingshu Δ Latency column is a *within-family, batched, serialized* cheap-vs-strong relative number (identical batch size and tensor-parallel degree), **not batch-1**; it moves the same direction as FLOPs where escalation is low (PMC −33%, SLAKE −22%) but goes *positive* where escalation is high (VQA-RAD +21%, MedXpert +60%), because the extra cheap pass is paid on nearly every query. The InternVL3 family and all PathVQA rows were measured under GPU contention and are **uncitable for latency** until a matched serial re-run (§10.5); their accuracy and FLOPs are unaffected.

Publication-standard efficiency summaries (APGR / CPT). We also report the two standard cascade summaries used in the deferral literature — **CPT** (compute-per-task, in 7B-forward-equivalents, *lower is better*; exactly $1 + \text{esc} \cdot 4.57$, so it also fixes the FLOPs column as $1 - \text{CPT}/4.57$) and **APGR** (an accuracy-gain-rate summary, *higher is better*; APGR > 1 marks a favorable accuracy-per-compute trade):

benchmark	APGR	CPT	ΔFLOPs
Lingshu (mean PMC-VQA APGR 1.225)	2.05	1.41	−69%
SLAKE	1.23	2.01	−56%
VQA-RAD	0.96	3.79	−17%
MedXpert	0.66	5.34	no win
MedVLThinker (mean PMC-VQA APGR 0.915)	1.32	2.33	−49%
VQA-RAD	1.08	2.69	−41%
SLAKE	0.72	—	no win
MedXpert	0.55	—	no win

APGR crosses 1 exactly at the win/no-win boundary (Lingshu: PMC/SLAKE > 1 win, VQA-RAD 0.96 marginal, MedXpert 0.66 loss), and CPT crosses 4.57 there (MedXpert CPT 5.34 > 4.57 = costlier than always-strong).

Threshold honesty. The matrix and per-benchmark tables report savings at the **iso-strong operating point** (τ chosen so cascade accuracy just meets the strong model). That is mildly optimistic, so we ran the honest version too: refitting τ on a **held-out split** and reporting the saving it *delivers*. On PMC-VQA the honest, held-out- τ saving is **−57% FLOPs for Lingshu** (cascade $0.563 \geq 32\text{B } 0.549$) and **−49% for MedVLThinker**, versus the fully-oracle- τ figures of −74% and −51%. So the headline PMC-Lingshu number reads three ways depending on protocol — **−69%** at the iso-strong operating point, **−57%** under a fair held-out threshold, **−74%** under oracle- τ — and the deployable, honest figure is the middle one, still a large saving. This is a threshold-protocol difference, not a contradiction; we report all three.

Gate-signal ordering. As in the ACC, the *choice* of confidence signal barely matters — the efficiency comes from the structure, not the gate. Among the three cheap signals, the ordering is **margin > confidence > cumulative-logprob**, with margin giving the lowest CPT at every benchmark: PMC-VQA margin CPT 1.41 (−69%) < conf 1.55 (−66%) < cum-logprob 1.64 (−64%); SLAKE margin 2.01 (−56%) < conf 2.06 < cum-logprob 2.19; VQA-RAD the three collapse (margin 3.79, −17%, $\approx \text{conf} \approx \text{cum-logprob}$). The spread is small precisely because the gate is saturated (§5–§6); margin is chosen as the marginally-best, training-free default.

10.4 The think tier: adding reasoning where it pays (the mirror image of over-thinking)

The ACC's premise (§4) is that reasoning **over-thinks** perception, so its no-think tier *removes* wasted thinking on the COMPETENT-4 sets. The complementary structural move is to *add* thinking exactly where it is under-supplied — the **reasoning** benchmarks — by running the strong leg in reasoning mode as a third tier. Faithfully, across all three families:

strong leg (reasoning mode)	MMMU-Med Δ	MedXpert Δ
Lingshu-32B	+0.034 (0.633 → 0.667)	~0 (floor)
MedVLThinker-32B	+0.107 (0.613 → 0.720)	+0.045 (0.299 → 0.344)
InternVL3-38B	+0.120 (0.633 → 0.753)	(<i>cap gap, fixup</i>)

Reasoning on the strong leg **adds accuracy on MMMU-Med for every family** (+0.034 / +0.107 / +0.120), and lifts MedVLThinker on MedXpert (+0.045, 0.299 → 0.344), while doing nothing on Lingshu-MedXpert (both models are at the near-chance floor there, so there is no headroom to recover). The behavior is thus **regime-adaptive**: think where headroom exists (MMMU), skip it at the floor (MedXpert).

The gains are genuinely from *more reasoning*, not prompt luck. On a generic "reason step by step" prompt all three families expand their decode from ~3 tokens of direct answering to real chains: **gen_toks 3 → 275 / 561 / 368** (direct → reasoning, per family). Notably, MedVLThinker emitted **0 <think> tags** yet still gained +0.107 — it does not need a native reasoning tag to benefit — which also **corrects an earlier retired claim** of ours that "Lingshu has no promptable think mode." A direct probe (lingshu_reason_probe.py) shows Lingshu-7B goes **gen_toks 3 (direct) → 174 ("reason step by step") → 267 (emitting real <think></think> tags)** on both MCQ and open-text; Lingshu *can* reason on demand, which is why the Lingshu-32B MMMU think-tier gain (+0.034) is available at all.

Finally, the think tier **composes into a FLOPs-positive cascade where the accuracy is there and stays away where it is not**. Gating the reasoning tier behind a cheap-tier check, MMMU reaches its think-mode accuracy at **~78%**

of **always-think FLOPs (a win)**, whereas MedXpert-Reasoning and MedXpert-Understanding land at **~143% and ~151% FLOPs (no win)** — the same regime-adaptive verdict, now paid for in compute: add the reasoning tier on MMMU, keep it off MedXpert.

10.5 Provisional and in-progress (honest status)

Three items in this section are explicitly **not final** and are excluded from the headline:

- **InternVL3-38B MedXpert-MM — not measured (cap gap).** MedXpert prompts run to ~20k tokens, above the 16,384 `MAX_MODEL_LEN` we set for the 38B (a KV-cache fit at $tp=2$), so both the direct and reasoning MedXpert runs for InternVL3 failed to generate. Both the 2-tier cell and the think- Δ cell are therefore blank ("cap gap, fixup"); a re-run at cap 24,000 is queued. This is the only structural hole in the InternVL3 column.
- **OmniMedVQA (7th benchmark) — running.** The full Open-access split is **88,996 QA** (4-option MCQ, exact-match; 42 sub-datasets, of which RadImageNet is 57k $\approx$ 64%); the Lingshu paper evaluates the full set (7B 82.9% / 32B 83.4%), so full — not a sample — is the faithful choice. A parser bug (a mid-run `KeyError`) has been fixed and the full run is in progress; **no OmniMedVQA cascade result is claimed here yet.**
- **Contended latency — uncitable.** As noted in §10.3, the InternVL3 family and all PathVQA rows were timed under GPU contention; their `latency_s` is withheld pending a matched serial re-run (accuracy and FLOPs are unaffected). Within-family cascade latency remains a *fair relative* cheap-vs-strong number (identical batch and tp , serialized), but it is not batch-1 and not cross-family comparable.

Status snapshot: **6 of the 7 Lingshu-suite benchmarks are complete across all three families**; the above are the remaining edges being closed.

11. Discussion and Limitations

This section states plainly what in the paper is new and what is inherited, bounds every headline claim to the regime where it actually holds, and lists the measurements that are still provisional. The standing rule applies throughout: every figure below is verbatim from a real checkpoint, and where a value is not yet measured we say so rather than estimate it.

11.1 What is and isn't novel

We do not claim new routing signals or new reward-model machinery. Two of the three methods are established mechanisms applied to a new setting; the third contribution is the negative characterization that explains *why* those are the levers.

ACC's decision rule is not new; its structure is the contribution. The agreement/consistency gate that fires the think tier (Eq. 3) is the ABC cascade family, and the confidence-margin gate at tier 0 is Chow/MSP. What is new is the *object* being gated: a cascade over **compute configurations of one large model** — a large-model *no-think* intermediate tier inserted between the small model and the large-model *think* tier — motivated by the empirical fact that reasoning over-thinks perception (§11.2). The systems contribution is incremental-but-defensible, not a new gate.

The verifier's mechanism is GenRM lineage; its application and unification are the contribution. A generative outcome verifier for best-of-N selection is standard. What is new is (i) applying it to **open-ended medical VQA and grounding** rather than closed-form or MCQ tasks, (ii) **unifying free-text answers and bounding boxes** under one score head with a single label rule ($y = 1[\text{IoU} \geq 0.3]$ for boxes), and (iii) using it as a *constructive counter* to the Verification-Mirage claim — a small trained verifier does break the training-free selection floor here, even though training-free self-verification does not.

The genuinely new empirical object is the luck floor itself. The finding that a dozen training-free gate signals all hit the same recoverability ceiling (pooled pick-correctness AUROC ~ 0.5 -0.69), that training-free selection sits at or below a random-pick floor, and that the two frozen models fail together, is a characterization result. It is what tells a practitioner to stop tuning gates and instead change the structure or train a small verifier.

11.2 Honest scope of the ACC over-thinking premise

ACC's headline efficiency (latency $\sim 80\%$, FLOPs to $\sim 52\%$, energy $\sim 5\times$ at ALL-6 parity) rests on the premise that the large model's *no-think* mode is at least as accurate as its *think* mode on the target queries. That premise holds on **perception** benchmarks and must not be over-generalized:

Where it holds (COMPETENT-4: SLAKE, VQA-RAD, PathVQA, PMC-VQA). MedVLThinker-32B no-think $\geq$ 32B think: SLAKE **0.849 vs 0.764** (+0.085), VQA-RAD **0.853 vs 0.776** (+0.077), with PMC-VQA (0.551 vs 0.556) and PathVQA (0.661 vs 0.672) essentially tied. Here the slow think pass is pure overhead, so inserting the fast no-think tier is a free win.

MMMU-medical is competent but a reasoning set, so the mechanism does not apply. MMMU is not near-chance — 32B-think reaches **0.688** — but here thinking *helps*: no-think **0.624** < think **0.688**. The over-thinking premise is false on MMMU, so ACC's no-think tier does not save compute on it. This is exactly why the reasoning "think tier" (§11 faithful-cascade results) is the right structure for MMMU while the ACC no-think tier is the right structure for perception; the two are regime-adaptive complements, not a single universal rule.

MedXpert-MM is the genuinely near-chance set and is excluded from headline efficiency. Both models sit at or below the 4-option chance level (7B MedXpert-R **0.225**, MedXpert-U **0.256**), so no routing decision over it is meaningful. It is excluded from the ACC efficiency headline and from COMPETENT-4, but it still appears in the full ALL-6 tables for completeness (ALL-6 parity 0.5723; ALL-5 removes MedXpert, parity 0.6463).

ACC does not gain on every base family, but it degrades gracefully. On families where think mode *hurts* overall (QoQ-Med, Chiron-o1), the agreement gate correctly collapses the cascade to the cheap 7B leg (escalation 0%, guard **0.00**) rather than harming accuracy — a no-gain but no-harm outcome, not a failure. The mechanism only *pays* where the large no-think mode is genuinely competitive; MedGemma is a partial case.

11.3 The verifier: matches-to-modest-win, not a blowout

The strongest single claim — a small trained verifier lets a 7B beat the strong 32B/38B on accuracy — is real but must be reported with its ties.

Against the strong model it is matches-to-modest-win. On the canonical two-seed audit (two 70/30 splits, pooled-4), the verifier beats the same-split 32B by **+0.039, 95% CI [+0.010, +0.066]** on seed 0 (excludes 0), but only **−0.005** on seed 1 (a tie). The honest summary is that the verifier **matches / is competitive with** the 32B, average **+0.017** — not a uniform blowout. The full-set 3-family master table does show every cascade-best cell above the strong model (e.g. pooled Lingshu 0.421 vs 0.331, MVT 0.344 vs 0.277, IV3 0.255 vs 0.218), but the per-split robustness check is what governs the headline verb.

Against training-free selection it is robustly positive on both seeds. Over greedy first-sample it captures **+0.088** (seed 0) / **+0.066** (seed 1) of accuracy, while self-consistency $\approx$ greedy. So the defensible, seed-robust claim is "beats training-free selection," and the "beats the 32B" claim is seed-dependent (holds on seed 0, ties on seed 1).

The per-dataset spread is wide. Pooled gap-captured is 35–49% across the two seeds (mean $\sim$ 42%), but per-dataset it ranges from little-headroom sets to hard sets:

dataset	greedy $\rightarrow$ verifier	gap-captured
SLAKE	0.738 $\rightarrow$ 0.762	15.2%
VQA-RAD	0.519 $\rightarrow$ 0.611	45.5%
PathVQA	0.352 $\rightarrow$ 0.441	55.7%
Kvasir	0.282 $\rightarrow$ 0.405	58.4%

SLAKE has little to recover (greedy already 0.738); Kvasir has the most. The often-quoted "35–78%" is the *cross-output-type* range spanning free-text pooled (35–49%) up to MS-CXR boxes (78%), not a per-dataset free-text range.

Other-base transfer is not uniformly robust. Trained from scratch on a *different* backbone (MedVLThinker-7B), the verifier works on SLAKE (42% of gap, greedy 0.564 $\rightarrow$ 0.622) but **fails on VQA-RAD** (0.500 $\rightarrow$ 0.470, −20%; n=54, noisy), pooling to 25% — positive but weaker than Lingshu's 49%. In contrast, *cross-generator* transfer (a Lingshu-trained verifier scoring MedVLThinker answers) is stronger: SLAKE 49% / VQA-RAD 61%. The takeaway is honest: verifier quality depends on the training base and the low-n sets are unstable.

It lifts selection over a frozen generator; it is not a SOTA grounder. The verifier chooses among a *frozen* generator's own samples — it does not produce better answers or boxes. Absolute grounding IoU stays modest (SLAKE box 0.255, MS-CXR box 0.232); the win is that it recovers 40% (SLAKE) and 78% (MS-CXR) of the gap to *that generator's own* oracle, with the MS-CXR bootstrap gain **+0.191, 95% CI [+0.152, +0.232]** excluding 0. We do not claim to beat a purpose-trained SOTA detector in absolute IoU.

11.4 Measurement caveats

Latency and energy are calibrated batch-1, not a single end-to-end wall-clock. They are computed from measured per-tier batch-1 costs composed through the cascade cost model (Eq. 1); FLOPs are exact (analytic $2 \cdot N \cdot (P+G)$). This is the honest accounting, but it is a modeled batch-1 figure, not a production-throughput number.

Some July latency was measured under GPU contention and is not citable. The InternVL3 family rows and all PathVQA rows in the July faithful-cascade matrix were measured while the GPUs were contended, so their latency_s

is uncitable until re-run serialized and matched (accuracy and FLOPs are unaffected). Clean serial re-runs (iv3_8b, four PathVQA legs) are queued. Cascade `latency_s` is a fair *relative* cheap-vs-strong number within a family (identical batch size and tensor-parallel degree, serialized), but it is not a batch-1 figure.

Threshold optimism is bounded but present. The MCQ efficiency numbers are reported at both an iso-strong operating point and a held-out fair τ (e.g. Lingshu PMC-VQA -69% at the iso-32B point but -57% on a held-out τ); we flag the more conservative held-out figure as the deployable one.

11.5 In-progress items (do not treat as final)

- **OmniMedVQA (the 7th benchmark, full 88,996 open-access QA) is still running** with the fixed parser; there is no cascade result for it yet.
- **InternVL3-38B on MedXpert-MM is not yet measured** — both the 2-tier FLOPs cell and the think-tier Δ . MedXpert prompts run $\sim 20k$ tokens, above the 16,384 context cap used for the 38B; a re-run at cap 24,000 is queued. The corresponding cells are marked "cap gap, fixup," not estimated.
- Status snapshot: 6 of 7 of the suite are complete across the three families.

11.6 The binding limit: candidate quality (future work)

The clearest ceiling on the verifier is **not** the verifier — it is the pool of candidates it must choose from. Three diagnostics converge on this:

family	per-answer AUROC	selection efficiency	verifier-bo8 vs oracle@8
Lingshu	0.903	81% (2027/2514)	0.414 vs 0.513
MedVLThinker	0.913	82% (1154/1414)	0.339 vs 0.416
InternVL3	0.898	74% (1688/2277)	0.249 vs 0.337

The verifier already ranks single answers well (per-answer AUROC ~ 0.90), and a ranking-loss variant pushes that to **0.93** — yet selection accuracy stays flat at ~ 0.50 , because on recoverable questions the correct answer is a minority vote and the verifier picks it only 74–82% of the time. The larger lever is candidate *diversity*: pooling answers across the three model families raises oracle@8 by **+0.11–0.15** (VQA-RAD +0.150, Kvasir +0.138, RadImageNet +0.113), far more than any better selector on a single model's samples. So the productive direction for future work is not a stronger gate or a sharper verifier but **better candidates** — cross-model pools, higher N (oracle rises through @16/@32), and generators tuned for useful diversity. Selection can only recover what the candidate set already contains.

11.7 Why two levers, and not a better gate

All of the above reduces to one fact. **Between** frozen models, the fixable-vs-not signal is nearly absent — a dozen training-free gates hit the same recoverability ceiling (AUROC ~ 0.5 –0.69), and the strong model fixes only 6–10% of the small model's errors (26% where it is competitive), unpredictably — so the leverage is to improve the **structure** (ACC). **Within** a frozen model, which-of-N-is-right is real but not zero-shot-surfaceable (training-free self-verification $\approx$ chance; a trained verifier reaches AUROC 0.924, and a trained 7B verifier beats a zero-shot 32B verifier 0.403 vs 0.355) — so the leverage is a **little training**. Structure and a little training are the levers precisely because the frozen-model routing/selection signal is, at training-free effort, luck.

12. Conclusion

A medical-VQA cascade must make two decisions, and this paper is a study of both: **which** queries to escalate to a larger model (the *gate*), and **what** to run when we do escalate (the *action*). The central negative result — the **luck floor** — is that over a *frozen* model neither decision can be out-engineered with *training-free* signals. On the gate side, a dozen confidence, conformal, learned, recoverability, and self-verification signals all land in the same AUROC band (~ 0.5 –0.69) at predicting whether escalation will help; the error structure of the two models is genuinely correlated (ALL-6 recoverability $\phi = 0.372$, i.e. the strong model fixes only 37.2% of the small model's errors), and on the competent four benchmarks the two models fail together most of the time ($P(32B \text{ wrong} \mid 7B \text{ wrong}) = 0.584$). On the action/selection side, training-free best-of-N selection sits at a luck floor: on SLAKE-open a random pick already scores 0.720, self-verification (0.715) is *below* random, and the best training-free selector (32B listwise, 0.758) captures only 24% of the gap to oracle@8 (0.879), because the correct answer is a *minority* vote in 74–90% of recoverable questions. The recoverability wall (Jitkrittum, NeurIPS'23) is the theory; our numbers are its medical-VLM instance.

Yet **two levers give large, real gains, and neither is "a better gate."** They are the paper's two positive contributions.

Structure buys efficiency. The Adaptive-Compute Cascade (ACC) routes over *compute configurations* rather than over models, inserting the strong model's *fast no-think* mode as an intermediate tier (7B-nt@cap320 $\rightarrow$ 32B-**no-think**@cap320 $\rightarrow$ 32B-**think**@fullres) and exploiting that reasoning *over-thinks* perception VQA (32B-no-think $\geq$ 32B-think on the competent sets, e.g. SLAKE 0.849 vs 0.764, VQA-RAD 0.853 vs 0.776). At honest held-out parity

with always-32B-think, on real measured batch-1 costs: **ALL-6 latency 11.34 s → 2.27 s (−80%), FLOPs 100% → 52%, energy 6318.8 J → 1181.9 J (≈5×), and zero guardrail violations** (no benchmark worse than always-7B); **ALL-5 8.88 s → 0.44 s (−95%), FLOPs → 24.9%, energy → 172.8 J**. The efficiency is the *structure*, not the gate — every training-free gate lands in the same 49–62% FLOPs band — and the same lever transfers to a *faithful MedEvalKit* MCQ cascade that matches the 32B/38B at **−17...−69% FLOPs** across three families, plus a reasoning "think tier" that *adds* accuracy where headroom exists (MMMU-Med +0.034 Lingshu / +0.107 MedVLThinker / +0.120 InternVL3).

A little training buys accuracy. A small trained outcome verifier for best-of-N selection *breaks* the luck floor and, unlike any training-free selector, **beats the strong model across three families and a held-out OOD set** (pooled cascade-best vs strong: Lingshu 0.421 vs 0.331, MedVLThinker 0.344 vs 0.277, InternVL3 0.255 vs 0.218 — the InternVL3 win uses the *Lingshu-trained* verifier, i.e. cross-architecture transfer). For free-text answers it recovers 35–49% of the oracle gap over two seeds, and the same principle holds for structured outputs, recovering 40–78% of the box-grounding gap including on a real chest-X-ray benchmark (MS-CXR bootstrap +0.191, 95% CI [+0.152, +0.232]). It behaves as a **test-time-scaling method**: adding candidates and selecting with the verifier lets a 7B match a 32B. Finally, the routing ceiling is partly an **MCQ artifact** — the *same* confidence signal moves from AUROC ~0.6 in multiple-choice to ~0.87 open-ended — so medical-VLM cascades should be evaluated open-ended.

The honest scope is stated as sharply as the wins. Against the strong model the verifier is a *match-to-modest* win, not a rout: over two 70/30 splits it is +0.039 (95% CI [+0.010, +0.066], significant) on one seed and −0.005 (a tie) on the other, averaging +0.017; it is robust only relative to *training-free* selection (+0.066/+0.088 over greedy on both seeds). The per-dataset spread is wide (free-text SLAKE 15.2% → PathVQA 55.7% / Kvasir 58.4% of the gap; boxes SLAKE ~40% → MS-CXR ~78%), a from-scratch verifier on a weaker base is not uniformly robust (it fails VQA-RAD at n=54), and the method lifts *selection over a frozen generator* — it is not a SOTA grounder. ACC's over-thinking premise holds only on *perception* (COMPETENT-4); MMMU is competent-but-reasoning, where the think tier *helps* and the over-thinking cut therefore does not apply, and MedXpert is near chance for both models and is excluded from the ACC headline. Several items are explicitly in progress and must not be read as final: OmniMedVQA (the 7th benchmark, ~88,996 QA) is still running, InternVL3-38B on MedXpert-MM is unmeasured (a 16384-token cap gap, re-run queued at cap 24000), and some July latency cells (the InternVL3 family and all PathVQA rows) were measured under GPU contention and are uncitable until re-run matched (accuracy, escalation, and FLOPs for those rows are unaffected). The binding limit throughout is **candidate quality**, not the selector: a cross-model candidate pool raises oracle@N by +0.11–0.15, which is where the largest remaining headroom lives and the clearest direction for future work. We release ACC, the trained answer and box verifiers, the faithful MCQ cascade and think tier, and the full negative-result characterization so that both the wins and the luck floor can be reproduced end to end.

Reproducibility index

Every number in this paper traces to a checkpoint. Code is launched from the repo root (~/medvlthinker-imgdiff-compute); the canonical value ledger is results/cascade_methods/artifacts/master_data.csv (MCQ ALL-6/ALL-5) together with results/cascade_methods/docs/current/{GROUND_TRUTH_NUMBERS.md → VERIFIED_FACTS.md, OPENTEXT_MASTER_TABLE.md, MASTER_SUMMARY_2026-07.md, METHODS_MASTER.md, METHOD_MATH.md, METHOD_ACC.md, UNIFIED_METHOD_EXPERIMENTS.md}; the session narrative is in progress/progress_{June_17, June_20-22, June_24, June_25-26, June_29-30, July_01-02, July_03}.md.

Method I — ACC (structure / efficiency). src/cascade_methods/{acc.py, acc_v2.py, acc_v3_confgate.py, acc_v4_lowres_think.py} (the 3-tier cascade and its agreement/margin/conjunction gates); the over-thinking premise and the strong-leg per-config comparison in src/cascade_methods/{strong_leg.py, final_comparison.py, overthink_generalize.py} from src/labeling/run_32b_modes_vllm.py; the all-methods bake-off, cross-family sweep, and head-to-head in src/cascade_methods/{acc_allmethods.py, acc_compare.py, cascade_all_families.py, gate_compare.py, compare.py}; per-tier measured batch-1 latency/energy via src/cascade_methods/{latency_estimate.py, open_measure_latency_energy.py} and NVML power in src/labeling/nvml_power.py; the cost math (Eq. 1) in METHOD_MATH.md/METHOD_ACC.md.

The luck floor (the unifying negative). Outcome-structure and recoverability diagnostics src/cascade_methods/{diagnostics.py, strong_fixes_genuinely_unknown.py, knowledge_feasibility_bytype.py}; capacity-vs-action analysis and cross-family complementarity src/cascade_methods/{crossfamily_agree.py, peer_premise.py, peer_router.py, peer_router_img.py, vision_sensitivity.py}; open-ended selection luck floor src/cascade_methods/{select_eval.py, ground_analyze.py, open_cascade_analyze.py}; writeups results/cascade_methods/docs/archive_mcq/{RECOVERABILITY_IS_CAPACITY_BOUND, OPENENDED_SELECTION_LUCKFLOOR}.md.

Open-ended ceiling-break. Open-VQA generation src/labeling/{run_openvqa.py, run_openvqa_fewshot.py}; the MCQ→open gate hunt src/cascade_methods/{gate_search_open.py, open_gate_bakeoff.py}; judge src/labeling/run_judge.py (MedVLThinker-32B and a Claude-Sonnet-5 cross-check, κ 0.85–0.96, 100% exact-match anchor).

Method II — trained verifier (accuracy). Answer verifier src/training_methods/{run_lora_verifier_open.py, run_lora_verifier_ranking.py, clean_verifier_dump.py}; the ablations that make it non-circular and pin the active ingredient src/training_methods/{verifier_image_ablation.py, verifier_transfer_eval.py, verifier_scaling_curve.py,

cross_gen_verifier.py}; the 3-family beats-32B master table (no new inference, from saved artifacts) src/cascade_methods/open_verifier_cascade_table.py, with the gate bake-off and cost frontier in src/cascade_methods/{open_gate_swap.py, open_gate_efficiency.py, open_gate_heldout_tau.py, open_recoverability_gate.py, open_cost_frontier.py}. Box verifier src/training_methods/run_lora_box_verifier.py; grounding data src/labeling/{run_ground_slake.py, run_ground_mscxr.py}; verifier defn (Eq. 4/5), best-of-N argmax (Eq. 2), and gap-captured metrics in METHODS_MASTER.md.

Method III — faithful MedEvalKit MCQ cascade + think tier (structure, generalized). The faithful harness is the vendored MedEvalKit/ (isolated venv + Qwen2_5_VL wrapper + Lingshu weights + datasets_path=hf + use_vllm + TORCHDYNAMO_DISABLE, vllm 0.9.0.1); the internal NGC harness is *not* faithful. Runners runners/{run_full_matrix_medeval.sh, run_native_think.sh, run_omnimed_reruns.sh, run_regen_native.sh}; analysis src/cascade_methods/{lingshu_medeval_cascade.py, cascade_all_families.py, lingshu_deferral_apgr.py (APGR/CPT), compare_native_think.py, control_think_signals.py, final_3tier.py, lingshu_prompt_probe.py}; reproduction-fidelity, per-benchmark FLOPs/latency/escalation matrix, held-out- τ honesty check, and the think deltas in progress/progress_July_03.md + MASTER_SUMMARY_2026-07.md.

No number in this paper is fabricated; every figure is copied verbatim from real checkpoint output and traces to a checkpoint via VERIFIED_FACTS.md / GROUND_TRUTH_NUMBERS.md.